Republic of the Philippines
Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City

PAW FEEDS: An IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring

A Capstone Project Presented to the Faculty of

Department of Information Technology

College of Computing Studies

In Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Information Technology

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Approval Sheet

The Capstone Project attached hereto, entitled **"PAW FEEDS: An IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring"**, prepared and submitted by **JHUDE PATHRICK V. MADROÑAL, ANTONIO MIGUEL APOSTOL, JOHN PATRICK E. BUCOL,** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, is hereby **recommended for Oral Examination**.

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Abstract

The fundamental challenge that this study will address is the difficulty busy dog owners face in maintaining consistent feeding schedules and ensuring their pets receive the correct, individualized portions when they are away from home. Traditional feeders often fail to provide the necessary real-time confirmation that food has been successfully dispensed, leaving owners uncertain. This research will develop PAW FEEDS: An IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring. The system's general objective will be to deliver a reliable, automated feeding solution that enables truly remote management over the internet, ensures precise portioning based on veterinary standards (RER/MER), and integrates pet-specific recognition. The methodology will utilize a Descriptive Developmental Research Design following the Agile development approach. The hardware will be structured around a dual-controller architecture: the ESP32-CAM (master) will manage high-level Wi-Fi communication and live video streaming, while the standard ESP32 (slave) will control the precise mechanical operations. The slave will be responsible for activating the MG996R servo-driven auger mechanism and handling the RFID/ISO15693 tag detection for individual pet identification. The system logic, developed in C++, will communicate directly with Firebase Cloud services via Wi-Fi for secure, real-time synchronization of commands, schedules, and dog profiles, allowing the React Native mobile application to manage the feeder from anywhere with an internet connection. Evaluation will include prototype testing (servo dispensing accuracy, RFID detection rate, and load cell monitoring) and a structured usability questionnaire administered to dog-owner participants.

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CHAPTER I

INTRODUCTION

Background of the Study

Feeding pets on time and getting the correct portion is a responsibility that many dog owners struggle to maintain, specifically those who are tend to go out and leave home for long hours. Consistent feeding time is important to our pet dogs, especially if we want them to be fed correctly. Missed or irregular feeding times may lead to problems such as overfeeding, underfeeding, or behavioral changes in dogs. The traditional methods such as manual scooping or even simple automatic feeders often fail to provide real-time confirmation that food has been dispensed, leaving owners uncertain about whether their pets have eaten or not.

With the little help of Internet of Things (IoT) technology, everyday routines such as pet care can now be automated and monitored remotely. Smart-automatic feeders provide opportunities for owners to schedule feeding times, automate food dispensing, and reduce the chances of meals being missed. However, many commercial feeders still lack accurate portion control and real-time monitoring features, which means owners cannot be certain if feeding has occurred or done precisely.

The specific problem that this study seeks to address is the absence of a system that integrates automated scheduling, precise portion dispensing, and reliable monitoring into one device. While existing feeders provide convenience, they do not confirm whether food was released successfully or measured according to veterinary standards.

This study introduces **PAWFEEDS**, an IoT-based smart dog feeder that integrates automation, monitoring, and portion control into a single system. The device consists of an ESP32-CAM (slave) for live video streaming and a conventional ESP32 (master) for scheduling, servo control, and REACT based mobile application. Dry food is dispensed precisely and reliably through an auger mechanism powered by a calibrated servo motor. As a supplementary feature, integrated RFID technology can recognize individual dogs using assigned tags. This allows the system to explore per-dog identification during feeding routines, while keeping its main focus on monitoring, automation, and usability. Owners can view feeding activity through live video, receive real-time alerts, and maintain confidence that feeding has occurred as scheduled.

Statement of the Problem

Dog owners face major challenges in feeding management:

1. Manual feeding is prone to human error or delays due to owner absence.
2. Existing automatic feeders dispense blindly without confirming if the dog approached or ate.
3. Owners lack real-time visibility (video/data) of the feeding event while away.
4. Missed meals or estimated scooping lead to overfeeding or underfeeding.

These problems often result in missed meals, overfeeding, or uncertainty for the pet owner. To address this, the system focuses on automated scheduling, accurate dispensing, and live monitoring. As an additional feature, RFID technology is integrated to explore the recognition of multiple dogs using individual tags.

Objectives

The general objective of this study is to develop an IoT-based smart dog feeding system that automates scheduling, ensures accurate portion control, integrates individualized pet recognition using individual RFID unique tags, and enables real-time monitoring and notifications through a mobile application.

Specifically, the study will:

- Design food dispensing mechanism for dogs.
- Create a React Android App for scheduling and alerts.
- Assess system accuracy (hardware) and usability (user acceptance).
- Construct a dual-dispensing mechanism with auger control.
- Incorporate RFID-based recognition as a supporting feature to identify multiple dogs using separate unique tags.
- Compare the system processes/performances to the traditional feeding
- Evaluate the system usability among the pet owners

Scope and Limitations

This study specifically focuses on the design and development of an IoT-based smart feeder that automates food dispensing, supports scheduling, and provides live monitoring features through a mobile application. The system ensures that dog owners can manage feeding routines consistently and remotely.

RFID functionality is included to recognize multiple dogs using individual tags. However, the dispenser is designed with a two-dispensing hole, meaning that food is released two portions at a time. While this allows per-dog identification and feeding, this limits the dispense for at least 2 dogs simultaneously.

Other limitations of the system include its compatibility only with dry dog food, since wet or semi-moist diets are not suitable for the auger dispensing mechanism. The accuracy of food portions may also vary slightly depending on the size and shape of the dry dog food, which can affect consistency in dispensing. The accuracy of the food portions also depends on the auger mechanism. Because of the design, it dispenses dog food by batch from each wing rather than by total weight. In addition, the system relies on a stable internet connection and continuous power supply for full functionality, as interruptions in connectivity or power may hinder notifications, monitoring, and scheduled dispensing. Finally, it is designed specifically for household use and is not intended for large-scale or commercial feeding applications.

Significance of the Study

The findings of this study will provide value to various stakeholders in the field of pet care and IoT-based solutions. The beneficiaries of this research include:

- **Dog Owners.** The system will provide pet owners the relief of not having to worry whether the dog has been fed or not. Live monitoring, RFID-based identification, and mobile notifications give owners confidence that each dog receives individualized care even when they are away from home.
- **Veterinary and animal nutritionists**. Portion calculations are based on veterinary portion formulas, the device follows science-based studies on calculating portions. This will help pets become healthier and active by maintaining this kind of proper diet.
- **Pet Care Industry.** The study offers a model for a cost-effective, IoT-enabled smart feeder that can be commercialized locally. The integration of RFID, camera monitoring, and mobile connectivity addresses gaps in existing commercial products, providing opportunities for innovation and business.
- **Researchers and Developers.** The system serves as a reference for future projects combining IoT, automation, and animal welfare. Its architecture demonstrates how dual ESP32 controllers, RFID modules, and mobile platforms can be integrated into a functional prototype.

- **Academic Community**. The research on dog feeding will provide more literature for future researchers who want to improve or adapt this study, which may serve as a key study in automation, computer engineering and future developments of IOT devices

By addressing real-world challenges in dog feeding, this study contributes not only to animal welfare but also to the advancement of IoT applications in everyday life.

Definition of Terms

This section provides the conceptual and technical definitions of terms that are essential to the study. The terms included here are specifically chosen because of their relevance to the system's design, development, and implementation. Defining these terms ensures clarity and consistency in understanding the concepts, components, and processes that make up the PawFeeds system.

Table 1 Definition of terms

Term	Definition
1. Internet of Things (IoT)	is a network of interrelated devices that connect and exchange data with other IoT devices and the cloud. IoT devices are typically embedded with technology, such as sensors and software, and can include mechanical and digital machines and consumer objects. [1]
2. ESP32-CAM	ESP32 is a chip that provides Wi-Fi and (in some models) Bluetooth connectivity for embedded devices – in other words, for IoT devices. While ESP32 is technically just the chip, the modules and development boards that contain this chip are often also referred to as "ESP32" by the manufacturer. [2]
3. Servo Motor	A servo motor (servomotor) is a highly specialized motor designed for precise control of rotary or linear motion. It's a rotational or translational motor that employs a feedback mechanism to ensure exact positioning, typically using a control signal that dictates the motor's movement to a desired position. This mechanism allows for precise control of various components, making servo motors crucial in applications where precise positioning and smooth motion are required. [3]

Term	Definition
4. Power Supply	A power supply is an electrical device that offers electric power to an electrical load such as laptop computer, server, or other electronic devices. The main function of a power supply is to convert electric current from a source to the correct voltage, current, and frequency to power the load. ()
5. load cell HX711 amplifier	The HX711 is a precision 24-bit analog-to-digital converter (ADC) designed for weighing scales and industrial control applications to interface directly with a bridge sensor. It is specially made for amplifying signals from load cells and reporting them to another microcontroller. [4]
6. ESP32 (Standard)	a chip that provides Wi-Fi and (in some models) Bluetooth connectivity for embedded devices – in other words, for IoT devices. [5]
7. Auger Mechanism	An auger is a screw-like mechanical device designed to move granular or powdered material through a rotating helical blade.
8. RFID (Radio-Frequency Identification)	Radio Frequency Identification (RFID) leverages radio waves to identify and track objects passively. Originally developed for commercial use in the 1970s, RFID technology has seen widespread adoption in recent years due to advancements in microchip and antenna technology, making it more accessible and affordable. [6]
9. React	Is a component-based JavaScript library used to build dynamic and interactive user interfaces. It simplifies the creation of single-page applications (SPAs) with a focus on performance and maintainability. [7]
10. Portion Control	Portion control refers to the process of dispensing food in precise, measured amounts according to dietary requirements. In our study we implement the veterinary standard RER.

Term	Definition
11. Caloric Requirement (RER/MER)	is a function of metabolic body size. Energy expenditure is related to surface area, and body surface area decreases as body weight increases. The equation for RER is the same for both cats and dogs. It is used to estimate the daily energy needed to sustain essential bodily functions (e.g. respiration, circulation, digestion, metabolism) while the dog/cat is at rest in a thermoneutral environment. These total daily energy requirements can then be ESTIMATED by multiplying the RER by an appropriate factor to estimate maintenance energy requirement (MER). [8]
12. Dual-Controller Architecture	A dual-controller architecture refers to a system design where two microcontrollers work together to share tasks. In this study, the ESP32 serves as the master controller managing feeding schedules, motor control, and RFID processing, while the ESP32-CAM functions as the slave controller handling video streaming. This division of labor improves reliability and efficiency, ensuring smooth operation of both dispensing and monitoring functions.
13. Firebase	The firebase Local Emulator is a set of advanced tools for developers looking to build and test apps locally using cloud Firestore, Realtime database, Cloud Storage for firebase, authentication, Firebase cloud functions(beta), pub/sub (beta), and firebase Extensions (beta). It provides a rich user interface to help you get running and prototyping quickly. [9]

CHAPTER II

REVIEW OF RELATED LITERATURE

Related Studies

Caring for our pets just got easier thanks to the help of technology today, especially managing their daily health care. This Review of Related Literature (RRL) will lay-out the scientific and technical groundwork for our project, PAW FEEDS: An IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring. In this chapter, we will look closely at existing research and systems that fall into three key areas: IoT technology used for remote control and scheduling; the mechanical engineering needed for accurate and precise food dispensing; and the use of RFID tags to identify individual pets.

Our project aims to fill a gap to the existing automatic feeders today. They often don't accurately measure food portions or give owners a reliable way to check that the food was actually dispensed. While many researchers have shown that basic microcontrollers like the ESP32 can handle scheduled feeding, very few systems combine all the crucial features into one reliable device: veterinary-standard portion control (using RER formulas), real-time video confirmation, and multi-pet recognition.

Therefore, the main purpose of this literature review is to find support for our design and to confirm that PAW FEEDS is a necessary and valuable addition to smart pet care, specifically by validating our hardware choices like the dual-controller (ESP32/ESP32-CAM) setup to keep the video stream from slowing down the critical dispensing actions, confirming the best way to dispense food using the auger-servo motor mechanism for accurate portions, and ultimately showing how PAW FEEDS provides a solution that is more reliable, confirmed, and personalized than current products. The following sections will present a focused analysis of the most relevant studies.

Nutrition Math 101: Important Calculation

[10] Highlighted the importance of accurate nutritional calculations in veterinary practice, particularly the use of RER and daily energy requirement (DER) to determine caloric intake for dogs and cats. The study explained that while the simplified formula (30 × BW + 70) is widely used, it may be inaccurate for pets under 2 kg or over 45 kg, leading to miscalculated feeding portions. Instead, formulas based on metabolic body weight (70 × BW^0.75) provide more precise results.

This aligns with the PAWFEEDS system, which incorporates RER- and MER-based formulas to automate portion control, ensuring pets are neither underfed nor overfed, thus supporting their overall health.

A Multifunctional Automatic Dog-Feeder with Bluetooth and Wi-Fi Connectivity

The increasing prevalence of household pets, particularly dogs, has led to a growing demand for efficient pet care solutions. Dogs, being the most common household pets, require regular feeding, hydration, and monthly medication to protect against pests. These tasks can become time-consuming for pet owners, especially those with multiple pets. To address these challenges, a study introduced a multifunctional automatic dog feeder designed to streamline pet care through automation and connectivity. [11]

The center of this all is the automated system known as ESP32 microcontroller, which enables many functions such as food dispensing, hydration, and medication administration. The system is designed to response to voice commands or button controls via the MIT app Inventor II, producing a user- friendly interface for pet owners. One of the standout features of this feeder is its ability to administer medication into the pets' food, making sure that the pets will receive necessary treatments without harming or giving stress to the pets.

In addition, the system includes three types of monitoring software Gmail, ThingSpeak, and a dedicated database website that will allow users to monitor

their pets' food and water levels remotely. System connection enables Bluetooth and WI-FI, confirming that the pet owners can remotely monitor and control the feeder from any location and at any time. The unification of these technologies determines the comprehensive solution that combines hardware and software components giving it an enhance pet care efficiency.

Development of an IoT-Enabled Automatic Poultry Feeder and Cage Temperature Monitoring System Using Microcontroller Technology

[12] Designed and implemented an IoT-enabled automatic poultry feeder and cage temperature monitoring system using a combination of Arduino Uno, ESP32, servo motors, relays, and the DHT11 temperature sensor. The prototype was capable of dispensing feed automatically based on scheduled times, while simultaneously monitoring and regulating cage temperature. If the temperature exceeded 30°C, the fan was automatically triggered, and if it dropped below 25°C, lights were activated. Feeding cycles were managed through a web server, and servo motors controlled the opening and closing of the feed container with high consistency. Testing demonstrated reliable performance, with WiFi-based IoT connectivity functioning within distances of 2 to 10 meters even with barriers.

The study emphasized the importance of automation in reducing labor time and preventing contamination of feed, which commonly occurs in conventional poultry farming. By integrating IoT, the system achieved improved feeding accuracy and environmental management, ensuring healthier and more productive poultry. This research supports PAWFEEDS by validating the use of ESP32 and servo motors for automated dispensing and by demonstrating how IoT frameworks can manage feeding schedules and environmental monitoring effectively. PAWFEEDS extends these principles by adapting them to domestic dog feeding, adding RFID recognition and live video streaming to personalize and monitor feeding sessions

Development of Fuzzy Logic Automatic Fish Feeding System and IoT-Based Water Quality Control

Developed an *IoT-enabled fuzzy logic-based fish feeding system with water quality control* for gourami aquaculture. The system integrated ESP32, temperature, pH, and turbidity sensors, along with servo motors to control feed dispensing. A fuzzy logic controller processed sensor inputs to regulate feeding duration, aerator operation, and temperature adjustments through heaters and coolers. Testing showed a very low error rate (<5%) across all sensors, and feeding accuracy had an error margin of only 0.19%, indicating precise and reliable operation. Notifications and monitoring were facilitated through Google

Spreadsheets and Telegram integration, allowing real-time feedback for farmers. [13]

For PAWFEEDS, it provides evidence that ESP32 combined with sensors and intelligent algorithms can deliver precise feeding outcomes, validating the system's choice of ESP32 for microcontroller operations. Unlike Indrawati's aquaculture system, PAWFEEDS extends this approach to the domestic pet domain, integrating RFID for individual pet recognition and ESP32-CAM for live video monitoring

Intelligent Pet Feeder with Real-Time Camera Monitoring

[14] Developed an *"Intelligent Pet Feeder with Real-Time Camera Monitoring"* that combines a PIR sensor, ESP32-CAM, NodeMCU microcontroller, and a servo motor to ensure automated feeding with visual confirmation. The system activates the camera upon motion detection, verifies the presence of a pet, and dispenses food in controlled portions. Testing showed 92% motion detection accuracy, 95% recognition accuracy, and 94% dispensing precision, with high user satisfaction ratings. This work supports PAWFEEDS by demonstrating the effectiveness of combining sensors and camera-based monitoring in pet feeding systems, though PAWFEEDS extends functionality with RFID for pet-specific recognition and container weight monitoring

Smart Pet Feeder (Application of Servo Motors in Automated Pet Feeders)

[15] Developed a smart pet feeder using an ATMEGA32 microcontroller, where a servo motor was employed to regulate the release of food from a PVC pipe into a bowl. Their study highlighted that servo motors provide the precision and controllability necessary to dispense measured portions of pet food. Unlike simple DC motors, servos allow controlled angular movement, making them suitable for timed and accurate food release in automated systems. The inclusion of the servo in their prototype demonstrated its effectiveness as a dispensing mechanism for small-scale feeders.

The PAWFEEDS system builds on this evidence by utilizing the MG996R servo motor, a widely used model known for its high torque and stability. While [15] used a smaller setup to validate the role of servos in feeders, PAWFEEDS applies the MG996R to handle larger and more consistent portions of dry dog food. The MG996R's durability and strength make it capable of rotating a dispensing mechanism reliably over repeated use. By integrating the servo with an ESP32 controller, PAWFEEDS ensures precise, scheduled dispensing, confirming that servos remain one of the most practical solutions for automated feeders.

Sigma-3: Integration and Analysis of a 6 DOF Robotic Arm Configuration in a Rescue Robot

Introduced Sigma-3, a rescue robot that uses a 6-DOF robotic arm powered by six high-torque MG996R servo motors. The study highlights the suitability of the MG996R for robotic manipulators due to its ability to provide sufficient torque and stability for lifting and rotating arm joints under varying payloads. The experimental evaluation demonstrated that the MG996R achieved precise joint rotation with less than 3% error in oscillation damping while maintaining low power consumption [16] These findings validate the servo motor's capacity for heavy-duty, repetitive motion tasks in robotics.

In the PAWFEED system, the MG996R is utilized to drive the 3D-printed auger mechanism responsible for pushing dog food portions from the storage container into the feeding tray. Since auger-based feeders require consistent torque to handle variable food weight and resistance, the MG996R's high-torque capability ensures reliable dispensing without jamming. This application aligns with the evidence provided by Bindu *et al* [16]., confirming that MG996R is not only suitable for robotic arms but also effective in automated feeding systems requiring precise, torque-driven actuation.

Design and Implementation of a Mobile-Controlled IoT Smart Pet Feeder For Busy Pet Owners

[17] Proposed PawTector, a smart feeding system that uses IoT technology and a smartphone application to enable automate feeding and track pet health in real-time. The method was created in response to the challenges experienced by busy pet owners who struggles to keep consistent feeding schedules. Their system includes a controlled mobile feeder that dispenses food and water using an ESP8266 a microprocessor, ultrasonic sensors, and a servo motor, as well a mobile application for schedule feeding and manual control feeding. Users can examine current food levels, track consumption habits, and create pet profiles that includes vaccination and care reminders.

The relevance to the PAWFEED system is clear in many ways. Both systems are intended to aid owners who are not always around and busy, the system provides remote feeding control and schedules dispense of food. The PAWFEED uses a camera instead of a sensor which is the ESP32-CAM for visual monitoring allowing owners to watch live their pet and feeding tray in real-time rather than depending on the sensor data. This not only decrease the system complexity, but to deliver more detailed feedback via live.

Surveillance Camera Using Wi-Fi Connection

Designed a Wi-Fi–based surveillance camera using the ESP32-CAM microcontroller and tested several video streaming protocols including HTTP, RTP, RTSP, and WebSocket. Their results highlighted that WebSocket delivered the smoothest real-time video with lower power use, while HTTP also provided a functional option but with higher resource demand. Despite WebSocket being

more efficient, the study confirmed that the ESP32-CAM is fully capable of supporting multiple streaming modes for continuous surveillance applications. [18]

In the context of PAWFEEDS, the ESP32-CAM is adopted for the live monitoring module so owners can observe feeding sessions in real time within the React mobile app. While the current implementation utilizes HTTP streaming instead of WebSocket, this prior research demonstrates that the ESP32-CAM remains a feasible and reliable choice for integrating camera-based monitoring into IoT systems. Thus, PAWFEED leverages the proven versatility of ESP32-CAM to provide pet owners with visual assurance during food dispensing, reinforcing both system reliability and user confidence.

ESP32-CAM as a programmable camera research platform

Presented the ESP32-CAM as a versatile, programmable camera platform for IoT applications. The module combines a 2MP OV2640 camera, dual-core processor, Wi-Fi/Bluetooth connectivity, and support for the Arduino environment, making it both low-cost and highly customizable. Their work demonstrated the ESP32-CAM's feasibility across multiple research projects, including synchronized multicamera systems (KameraflY), interchangeable-lens cameras (KISS-E), and large-format scanning cameras (Lafodis). Key findings emphasized its programmability, real-time live view capability, local storage options, and wireless synchronization for coordinated camera tasks [19]

For the PAWFEED system, this study strengthens the decision to utilize ESP32-CAM for the live monitoring module. While PAWFEED uses a simpler setup with HTTP streaming instead of advanced synchronization or custom optics, the research highlights that ESP32-CAM is robust enough for experimental and production-level IoT camera systems. This proves its suitability for continuous pet feeding surveillance, providing owners with reliable visual confirmation during automated dispensing.

Design and Development of a Smart Pet Feeder with IoT and Deep Learning

[20] Created a smart pet feeder that combined IoT hardware with deep learning to enhance feeding automation. Their design used sensors, a dispensing mechanism, and a camera module to identify pets through trained recognition models. The study emphasized that integrating artificial intelligence can improve feeding accuracy by ensuring the right animal is served.

PAWFEEDS relates to this work by sharing the goal of reliable and accurate monitoring, but it takes a different approach in terms of recognition. Instead of relying on deep learning and computer vision, PAWFEEDS uses RFID technology to identify individual dogs during feeding sessions. This offers a simpler and more cost-effective solution, as RFID is less resource-intensive and works reliably regardless of lighting conditions. At the same time, the ESP32-CAM in

PAWFEEDS is used primarily for live video confirmation, giving owners real-time assurance of food dispensing rather than handling recognition.

Automatic Pet Feed System Applying IoT

This study developed a smart pet feeding system based on the Internet of Things (IoT) using an ESP32 microcontroller, load cell, ultrasonic sensor, and DC motor with an auger to dispense food. The system included a mobile application using On-Device Storage, enabling real-time monitoring of food intake and storage levels while allowing customization of feeding schedules according to the pet's body condition, breed, and weight. Accuracy testing showed the IoT feeder could dispense consistent food portions, ensuring pets remained within their healthy weight range. While effective, the system relied on generic cloud integration and lacked RFID-based identification or multi-pet management. In contrast, PawFeeds expands this model by combining RFID tag recognition, live ESP32-CAM monitoring, and MER/RER-based nutritional computation in a custom React application [21]

Enhancing IoT Based Aquamonitor Using React.js and Node.js Framework

[22] Developed an IoT-based aquaculture monitoring system utilizing React.js for the front-end **and Node.js** for the back-end. The system monitors real-time water parameters such as pH and temperature, displaying them through an

interactive React dashboard. This approach showcases how modern web technologies can improve data visualization and user experience in IoT systems. The responsive interface ensures that users receive timely updates and alerts, demonstrating how IoT data can be managed and presented efficiently. For the PAWFEEDS system, this study provides a foundation for integrating interactive interfaces with IoT hardware. Although the aquaculture system uses React.js for web-based monitoring, the same design concept applies to React Native, which PAWFEEDS utilizes for its mobile interface. This ensures users can manage feeding schedules and monitor dispenser activity in real time through a mobile app with smooth and responsive performance.

Design and Implementation of an Open-Source IoT and Blockchain-Based Peer-to-Peer Energy Trading Platform Using ESP32-S2, Node-RED, and MQTT Protocol

[23] Introduced an IoT-enabled energy trading platform combining ESP32-S2 microcontrollers, Node-RED, and React.js for system monitoring. The architecture integrates blockchain to secure data transactions, while React.js provides a dynamic interface for visualizing energy flow and peer-to-peer exchanges. This combination of IoT hardware and modern web technologies highlights the flexibility of React in handling real-time sensor data and secure communications between devices.

In relation to PAWFEEDS, this study emphasizes the effectiveness of combining IoT with a robust and interactive interface for control and monitoring. While the original research used React.js for a web application, PAWFEEDS adapts the same IoT principles using React Native to provide mobile access and real-time communication between the app, Firebase, and the ESP32 controllers, ensuring accurate and reliable feeding automation.

A Fully Integrated RFID Reader SoC

Designed a fully integrated RFID Reader SoC that supports multiple high-frequency standards, including ISO/IEC 14443 Type A, Type B, and ISO/IEC 15693. Among these, ISO/IEC 15693 is especially significant because it operates at 13.56 MHz and allows longer communication distances of up to 1 meter, compared to ISO/IEC 14443 which is limited to near-field ranges of about 0–10 cm . The study confirmed successful communication with ISO/IEC 15693 tags using 10% ASK modulation for inventory commands and Manchester encoding with load modulation for tag responses, validating its compliance with the standard [22].

For the PAWFEED system, which aims to regulate pet feeding through RFID-enabled identification, ISO/IEC 15693 provides a strategic advantage. Its ability to support vicinity cards ensures that dogs wearing RFID tags can be identified reliably even at greater distances or while in motion around the feeder.

This reduces the risk of missed detections and enables multi-tag identification, which is crucial when multiple dogs are present. By planning to utilize ISO/IEC 15693-based RFID, PAWFEED aligns itself with a robust and widely adopted international standard, ensuring both scalability and accuracy in automatic feeding applications.

Assessing the Validity and Reliability of A Low-Cost Microcontroller-Based Load Cell Amplifier for Measuring Lower Limb and Upper Limb Muscular Force

Evaluated the HX711 microcontroller-based load cell amplifier as a low-cost and portable alternative to commercial-grade systems for measuring muscular force. Their study compared the HX711 with both a standard laboratory signal conditioner and a handheld dynamometer, reporting nearly perfect correlations (Pearson's R between 0.947–0.992) and strong reliability scores (ICC values above 0.90 for both inter-day and intra-day tests). These findings confirmed the validity and reliability of the HX711 in accurately measuring isometric muscular strength [25]

The HX711 amplifier works by converting the small voltage signals from a load cell into digital values that can be processed by a microcontroller such as an Arduino. It is highly customizable in terms of calibration, sensitivity, and measurement ranges, making it adaptable to different contexts [25]

For the PAWFEED system, the HX711 amplifier is not primarily used for precise portion-by-portion weighing but rather for monitoring the container's food status. Its role is to detect whether food is present or depleted, ensuring that owners are notified when the dispenser requires refilling. This application takes advantage of the HX711's reliability, small size, and low power consumption, making it well-suited for continuous monitoring in an IoT-based feeder. By integrating HX711, PAWFEED prevents interruptions in scheduled feeding due to an empty container, thereby supporting the system's overall dependability

Synthesis

The reviewed studies consistently emphasize the importance of automation, accuracy, and monitoring in animal feeding systems. Several works focused on servo-driven dispensing, confirming its reliability for portion accuracy in both small-scale and larger feeding mechanisms. Others expanded on this by integrating microcontrollers such as ESP32 and ESP8266, enabling IoT-based scheduling and remote monitoring. Camera-based systems demonstrated the feasibility of using ESP32-CAM modules for real-time visual confirmation of feeding activities, while advanced approaches incorporated artificial intelligence for pet recognition. Despite these contributions, limitations remain most systems either rely solely on sensors, lack per-animal identification, or do not unify multiple features into a single solution. To address these gaps, PAWFEEDS combines IoT scheduling, servo-driven portioning, RFID-based identification, and ESP32-CAM monitoring into an integrated multi-module feeder, ensuring reliable, precise, and individualized pet feeding

Table 2 Synthesis

Study Title	Hardware / Platform Used	Key Features	[4]	Proposed study
A Multifunctional Automatic Dog-Feeder with Bluetooth and Wi-Fi Connectivity [11]	ESP32 + Servo + Wi-Fi/Bluetooth	Automates dog feeding, hydration, and medication; remote monitoring via Gmail/ThingSpeak	No RFID, no camera integration; limited to single pet	PawFeeds adds RFID-based recognition, ESP32-CAM live monitoring, and MER/RER-based portion computation
Development of an IoT-Enabled Automatic Poultry Feeder and Cage Temperature Monitoring System [12]	**ESP32 + Arduino + Servo + DHT11**	Automated poultry feeding and cage temperature monitoring; web server scheduling	Poultry-specific, lacks RFID and live monitoring	PawFeeds adapts scheduling and dispensing to dogs, adds RFID, camera, and individualized feeding
Development of Fuzzy Logic Automatic Fish Feeding System and IoT-Based Water Quality Control [13]	ESP32 + Sensors (temp, pH, turbidity) + Servo	Fuzzy logic-controlled fish feeding; IoT-based water quality monitoring; Telegram alerts	Designed for aquaculture, no RFID, no multi-pet support	PawFeeds applies IoT to companion pets with RFID identification and nutritional computation

Study Title	Hardware / Platform Used	Key Features	[4]	Proposed study
Intelligent Pet Feeder with Real-Time Camera Monitoring [14]	**ESP32-CAM + Servo + PIR**	**Automated feeding with live video monitoring; pet detection**	**No RFID or nutritional calculation; focused mainly on camera**	PawFeeds combines camera monitoring with RFID-based identification, multi-pet profiles, and MER/RER formulas
Design and Implementation of a Mobile-Controlled IoT Smart Pet Feeder for Busy Pet Owners [17]	**ESP8266 + Servo + Ultrasonic**	**IoT-controlled pet feeder with mobile app scheduling and reminders**	**No RFID, no camera integration; generic mobile interface**	PawFeeds enhances with React custom app, RFID scanning, and ESP32-CAM monitoring
Automatic Pet Feed System Applying IoT [21]	**ESP32 + Load Cell + Ultrasonic + DC Motor (auger) + Firebase**	**Dispenses accurate portions; monitors food intake and storage levels; mobile app integration**	**No RFID, no live camera; limited personalization**	**PawFeeds extends this with RFID recognition, live ESP32-CAM streaming, and multi-pet management**

Conceptual Framework

The conceptual framework illustrates the interaction between the user, mobile application, local database, and hardware components of the PAWFEED system. At the presentation layer, the user interacts with the mobile application to set feeding schedules, trigger manual feeding, register dog profiles, and monitor feeding activity. The mobile application, developed in REACT, communicates with both the database and the hardware, sending commands, retrieving data, receiving RFID-verified feeding logs, and displaying notifications, food storage levels, and live camera feeds. The database serves as the central repository, storing feeding schedules, dog profiles linked with RFID tags, and the food container status measured by the HX711 load sensor. This ensures that feeding events are recorded, dog-specific portions are managed accurately, and container levels are tracked for timely refilling. Firebase acts as the cloud backend database service for the PAWFEED, this serves as the bridge between the mobile application and local database itself. Through firebase authentication, users can deliberately secure login, update feeding configurations, and receives notifications. This allows to enhance the scalability, allowing users to access the system while reducing the dependency on the local-storage.

The hardware layer integrates the ESP32 Devkit (master), RFID/ISO15693 module, HX711 load cell, and ESP32-CAM (slave). It retrieves feeding schedules from the database, scans RFID tags to verify the correct dog before dispensing, validates food levels, executes automated or manual dispensing, and streams live camera feeds. Notifications and feedback are sent back to the mobile app for user monitoring and control.

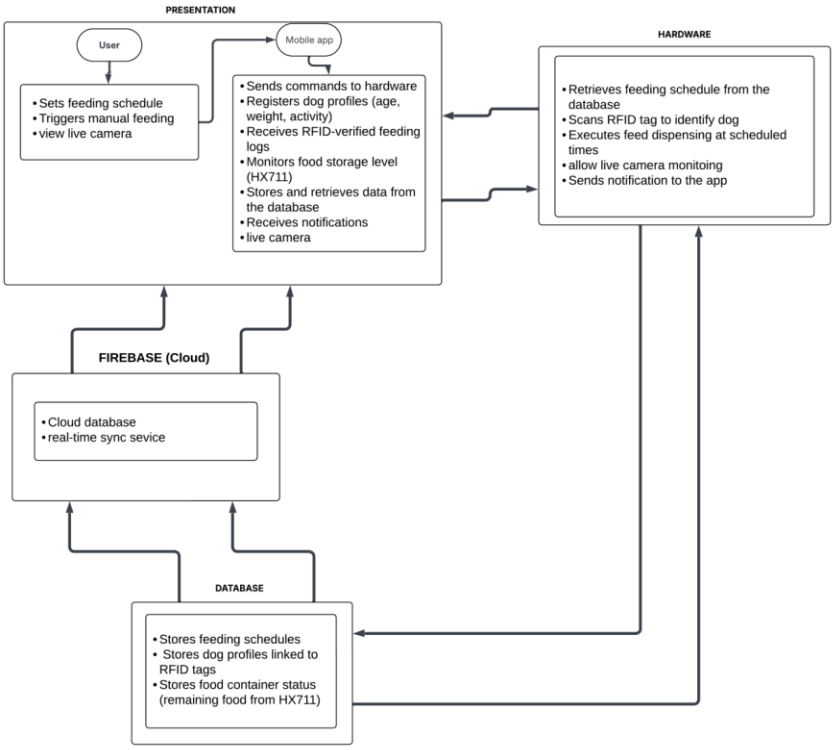


Figure 1 Conceptual Framework

CHAPTER III

METHODOLOGY

Research Design

This study applies a Descriptive developmental research design, that combines the systematic development of a technological product with the documentation and analysis of its performance and usability. The objective is to design, develop, and assess the effectiveness of PAWFEEDS: An IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring that supports remote feeding control through scheduled and manual triggers.

The descriptive component of the study focuses on observing and analyzing the behavior of the system during developmental and testing specifically how well it performs its intended, in this case the function that dispense pre-measured food portions at scheduled times and receives signal from the RFID unique tags to communicate to the system. It includes the usability and reliability of the mobile application interface, making pet owners manage feeding routines and scheduling via WI-FI connectivity.

This study focuses on developing the system with continuous refinement and improving both hardware and software. At the center of the operation is the ESP32devkit processor, that controls the calibrated servo motor for scheduled feeding time. Feeding schedules are to be set through the REACT based mobile application that will provide the interface for scheduling, monitoring, feeding and control. As an additional supplementary feature, the RFID is used to identify individual dogs via RFID unique tags.

The system operates in a simple, decision-based rule logic programmed in C++ using the Arduino IDE. the mechanism for dispensing is triggered when the current time matches the user's preset schedule, and recognizing RFID unique tags. Through this design, the researchers aim to determine if the system design which combines IoT, and real-time control can provide a practical and user-friendly solution for pet owners, particularly those who may be away from home or have very limited time to manage feeding routines manually.

Data Source

This study will be obtained from both primary and secondary sources to support the design, development, and evaluation of the PAWFEEDS system.

The primary data will be gathered from the prototype testing of the PAWFEEDS system. These will include logs such as RFID unique tags detection rates, calibrated servo motor dispensing cycles, load cell readings, and ESP32-CAM (slave) live video monitoring. The user feedback will include structured responses from the selected participants concerning of the usability of the mobile application and the feeder's performance.

The number of respondents will be chosen at least 30 respondents and from that we will randomly select 5 evaluators to conduct hands on to the device, to get to provide different perspectives on usability and functioning and remain on the project's limits. the criteria will only require respondents to at least own one dog and uses smartphones to gain access to the mobile application, while the exclusion criteria will exclude those individuals who do not have a WI-FI connection. While the sample size is modest and base on convenience, it will be sufficient for a prototype-level evaluation, where in the main goal is to test the system's operation rather than generalize to a larger population.

The secondary data will be gathered from technical datasheets, manuals, and related studies that will guide the development of PAWFEEDS. The RFID module datasheet will be used to check its own range, operating frequency and standards so that tag scanning will work accurately. The MG996R servo motor will be followed to know its torque, speed, and voltage needed for the auger to dispense food properly. While the HX711 load cell sensor will be used to make sure the weight readings will be accurate for portion control. The ESP32-CAM (slave) specifications will be reviewed to set up its camera, memory, and Wi-Fi features, for real time monitoring. Lastly, we have the veterinarian guidelines such as the Resting Energy Requirement (RER) and Maintenance Energy Requirement (MER) will be applied to compute the right food portions.

Data Gathering Instrument

The study will utilize a digital survey and evaluation tool administered via google forms to collect the data needed. The platform google forms was use to facilitate effective data collection from the 30 respondents regarding how their dogs and also 5 evaluators who actually tested the machine to give us their experience and suggestions.

Data Gathering Technique and Procedures

The PAWFEEDS system will be tested through controlled demonstrations designed to evaluate both technical performance and user interaction. The trial runs will include RFID scanning to identify tagged pets, servo-driven dispense of food portions, and verification of real-time video streaming via the ESP32-CAM (slave). Selected evaluators from the 30 respondents will who answers our survey first then select 5 evaluators from the respondents. They each use the app and device to set up schedules, monitor, and then filled out the evaluation form to rate their experience. Their feedbacks will be collected via google forms to easily record every answered questions. This combine approach will assure that the data gathered will be use comprehensively to reflect both the operational reliability of the prototype and the perceived effectiveness of the systems overall evaluations.

Ethical Considerations and Data Privacy

In accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), the PAWFEEDS system applies these strict measures to guarantee the security of the participants data and the data collected by the system.

The PAWFEEDS researchers are prioritized the security and anonymity of the respondents. Participants are provided with informed consent, accepting their involvement to the survey and testing and it is voluntary that they can withdraw at

any time. All data physical and digital records of the survey and testing are kept confidential and can be only accessible to the researchers.

The PAWFEEDS system is built by privacy-by-design. The system collects only minimal data and information of the user like the email for login, pet birthday/weight for feeding calculations that is strictly necessary for operating the system. The users are free of control to view or edit the data via app. Sensitive data like passwords, are encrypted in the database. Disposal of the account and the data depends on the user. Users have the right to request of removing the account, upon which all personal and pet data will be permanently removed from the system.

Operational Guidelines

By using the system, users are responsible for the accuracy of the pet information (weight, birthday) to ensure the pets feeding portions are safe. Users recognize that the PAWFEEDS system is an assistive tool but do not wish to replace professional veterinary advice. Users are also required to keep the confidentiality of their credentials to prevent unauthorized access.

Data Analysis

To easily understand the answers, we got from the google forms, we used a simple math to analyze the data. For part A of our survey, we use Frequency and percentage, since we have 30 respondents for the survey this help us count how many dog owners have more than one dog, what kind of food do they feed, and how often does their pets eat. By doing this we basically picture out who are the respondents. For the part B where only the 5 selected evaluators can rate the system, we use the weighted mean to rate the device and system 1-5. We calculated the average score for each question to see if they actually like the features. This is how we will see if the system was reliable and easy to use.

Software Development

The software development for the PAWFEEDS were divided into three distinct, the embedded firmware for the hardware, cloud-based for the backend structure, and the frontend user application.

The embedded firmware for the main controller, the ESP32-devkit, was developed in C++ using the Arduino IDE. The controller was programmed with decision-based logic to manage all the operations. It includes WI-FI connectivity, querying the database for feeding schedules, and monitoring the system clock. Upon a scheduled preset time, it initiates RFID tags validation, a successful scan can trigger the actuation of the servo motor and auger and also the load cell for precise portion dispensing. The controller transfers the operational logs and updates the data to the backend. At the same time, a separate firmware is deployed to the ESP32-CAM (slave) which turns it into a simple web server just for streaming the live video feed.

The backend system, which is built using Firebase. We used the firebase real-time database to implement as the centralized data repository. The backend is in charge of storing, retrieving, and synchronizing all critical data, which includes a user-defined schedules, dog profiles, portion sizes, and feeding logs. The cloud-based ensures that both the embedded hardware and the frontend application can access current data available in real-time.

The final part is the frontend, which was developed in REACT-based mobile application, serving as the primary interface for the system. Like any other interface, this application provides an interface for the feeding schedules, dispensing cycles, and monitors real-time feeding logs and food levels. In addition, it integrates the ESP32-CAM (slave) that shows the live streaming video, enabling users to access the real-time monitoring feature directly from the application.

Development Approach

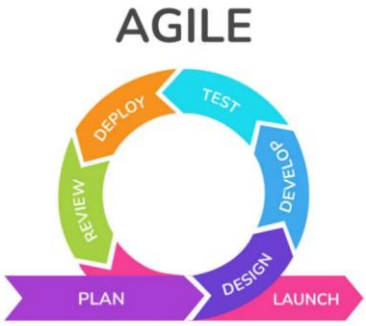


Figure 2 Developmental Approach

For the development of the PAWFEEDS automated dog food dispenser, the agile methodology will be adopted as the software development approach. Agile will be chosen due to its flexibility, focus on user collaboration, and ability to deliver working software in iterative cycles. This methodology will allow the development team to adapt to evolving requirements, apply user feedback quickly, and continuously improve both the hardware and software components of the system. The project will be divided into multiple short development cycles, each focusing on a specific feature set such as scheduling, food dispensing, or manual control

using mobile. At the end of each sprint, a working module will be delivered, reviewed, and refined base on the feedback from end-users. The initial sprints will focus on core functionalities such as settling feeding times, live camera monitoring, power optimization, activating the servo motor, and sending commands via REACT Mobile App. Later sprints will address the individualized RFID tags recognition, and user interface enhancements. Regular stand-up meetings and sprint reviews will be conducted to assess progress, identify roadblocks, and reprioritize task as needed. This iterative cycle of planning, development, testing, and feedback will ensure that PAWFEEDS evolves into a reliable and user-friendly IoT pet care system that directly addresses the needs of dog owners.

System Design

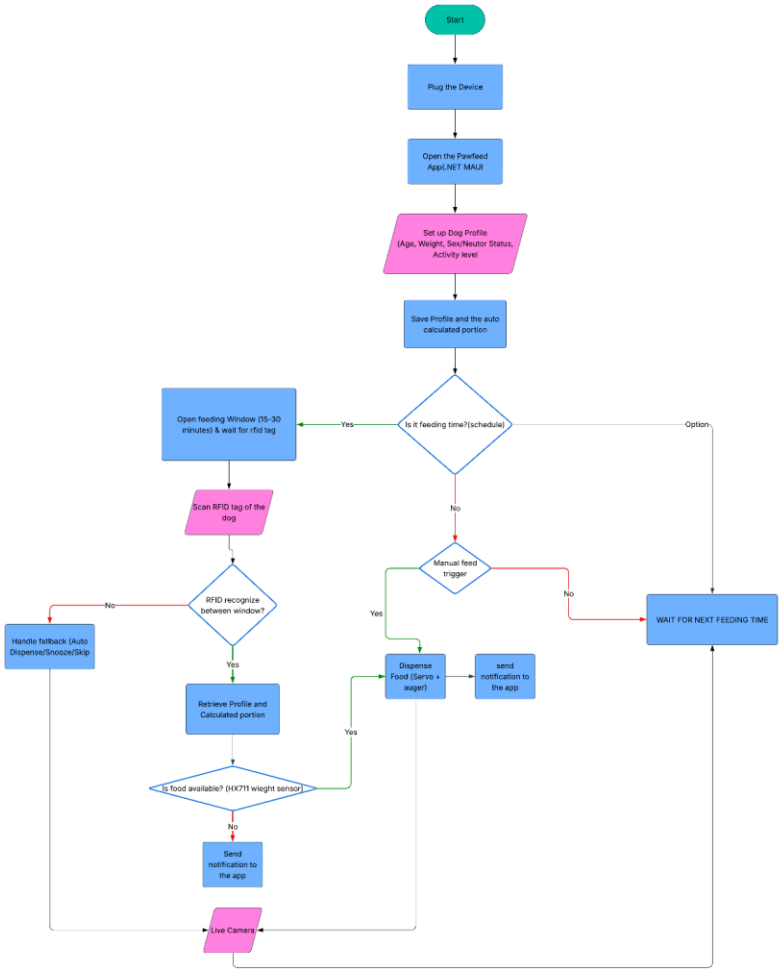
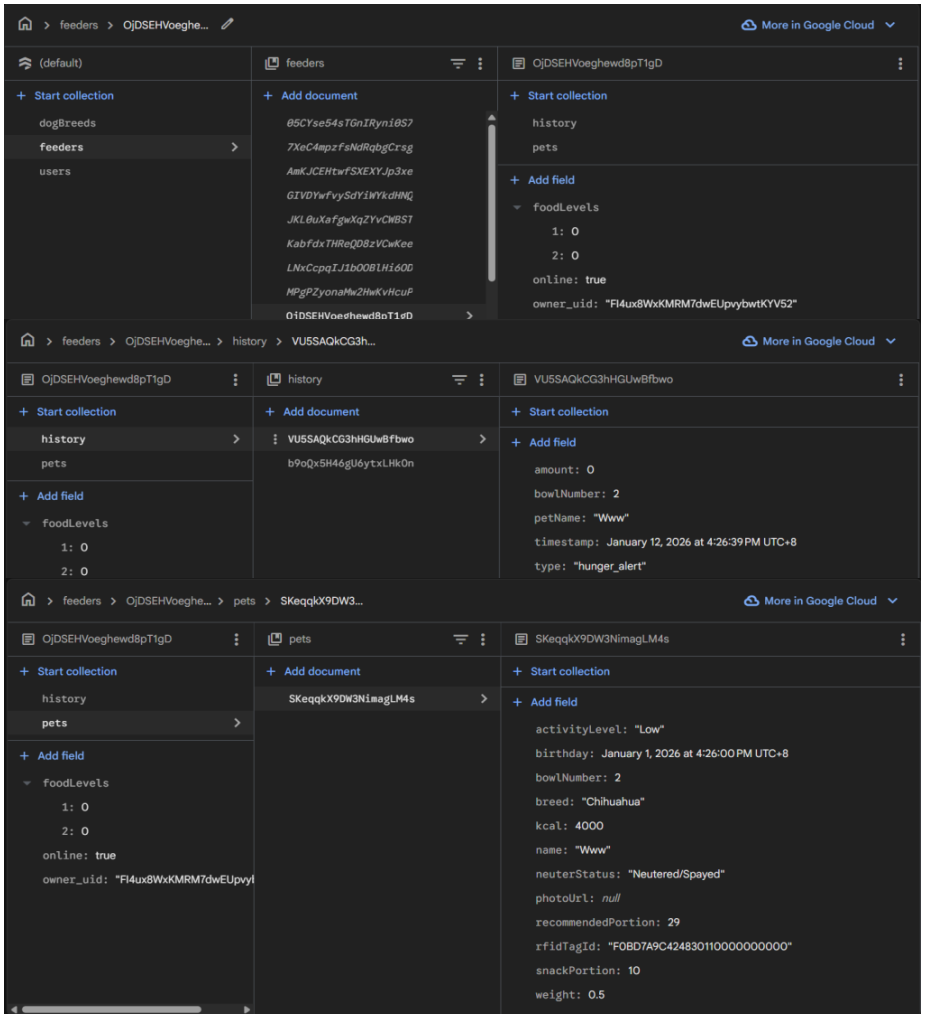


Figure 3 System Flowchart

system flowchart illustrates the step-by-step process of how the PawFeed system manages automated and manual feeding. Once the device is powered on, the user opens the PawFeed mobile application developed in REACT to register and save the dog's profile, which includes birthday, weight, sex/neuter status, and activity level. Based on this information, the system automatically calculates the appropriate food portion for each dog and the system wait for a RFID tag to be scan to officially add the RFID unique tag to that dog.

During feeding time, the system checks the schedule and opens a feeding window of 2 minutes, waiting for the RFID tag of the corresponding dog to be scanned. If the RFID is recognized within this window, the system retrieves the dog's stored profile and calculated portion. While dispensing, the HX711 weight sensor validates the dispense of food if it hits the total grams needed to be dispense, the auger will stop dispensing. While the ESP32-CAM (slave) provides a live feed and a notification is sent to the app. In cases where no RFID tag is detected, the system handles fallback such skip meal. Manual feeding can also be triggered through the app, which follows the same validation, dispensing, and notification process.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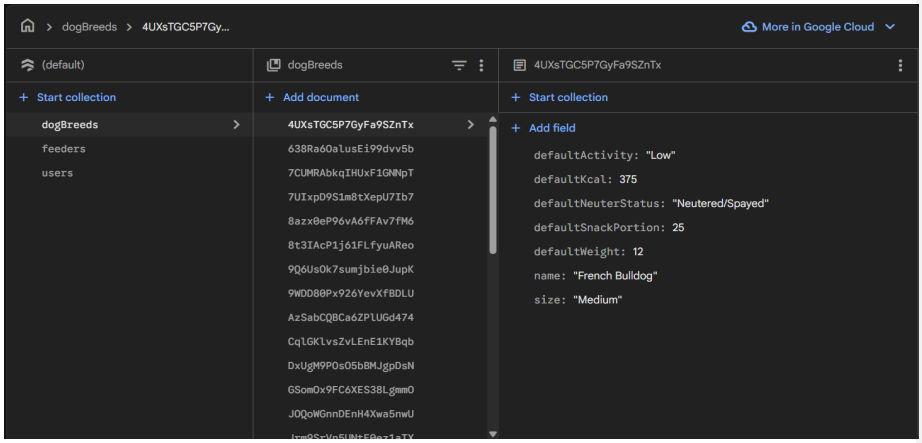


Figure 4 Database Schema

The main repository for persistent application data in the PawFeeds database schema is Google Cloud Firestore, which is arranged into three root-level collections to separate user identities, biological references, and hardware states. The users collection manages administrative privileges and authentication profiles while securely connecting to Firebase Auth UIDs to store user data and push notification tokens. The recommended food portions for pets are determined algorithmically based on breed-specific characteristics using the dogBreeds collection, a static reference database seeded with breed-specific physiological data such as metabolic rates, default weights, and activity levels.

Commented [JA1]: Database Schema must be the actual database of the system.

The core of the IOT architecture is the feeders collection, which acts as the digital equivalent of the physical hardware. Important state information such as camera stream availability, owner linkage, connection status, and current food container levels are preserved in each document in this collection, which each

represents a unique ESP32 device. The mobile application can check the hardware status asynchronously without requiring direct peer-to-peer connections between the feeder and the phone thanks to this collection, which serves as the primary hub for device management.

The system uses a nested sub-collection architecture in each feeder document to handle intricate data relationships. Individual animal profiles are stored in the pets sub-collection, which connects distinct RFID tag IDs to particular dietary requirements and milestone tracking logic. The schedules sub-collection manages feeding automation by specifying exact dispensing times, repeat intervals, and portion sizes that are monitored minute by minute by cloud functions. A history sub-collection additionally offers an unchangeable record of every dispensing event, facilitating the creation of user health analytics and consumption reports.

The system uses the Firebase Realtime Database only for high-speed, temporary command execution in parallel to Firestore. Between the cloud backend and the ESP32 microcontroller, this independent service manages the instantaneous transmission of commands, like await_rfid commands, and reports hardware error flags, such as jams or empty hoppers. This division of responsibilities guarantees that the latency-sensitive hardware commands are carried out with the least amount of delay while the persistent user data in Firestore is kept organized and queryable.

Functional Requirements (System Features)

Requirement No.	1
Event	ESP32-CAM via live feed monitoring
Description	The ESP32-CAM monitors the dispensing of dog feeds.
Rationale	To provide visual on the dog bowl.
Source	(Apostol, Madroñal, Bucol)
Fit Criterion	The ESP32-CAM must stream live video to the mobile app and a frame refresh rate of at least 2 frames per second under stable WI-FI.

Requirement No.	2
Event	Weight measure via load cells
Description	Use to measure the weight of food being dispense.
Rationale	Ensures accurate food dispensing, prevents overfeeding, underfeeding, and keeps feeding consistent.
Source	(Apostol, Madroñal, Bucol)
Fit Criterion	The load cell starts detecting the weight of the bowl once the dispensing starts.

Requirement No.	3
Event	HX711 (load cell amplifier)
Description	The HX711 module will measure the weight of the dispensed dog food through the load cell sensor.
Rationale	Accurate weight measurement is needed to ensure the dog receives the right portion of food.
Source	(Apostol, Madroñal, Bucol)
Fit Criterion	The recorded values should match the actual weight of the dispensed food when tested multiple times under controlled conditions.

Requirement No.	4
Event	Servo Motor MG996R (Auger Mechanism)
Description	The MG996R servo motor will rotate the auger to dispense precise portion of dry dog food.
Rationale	Using a servo motor ensure that the auger can handle the resistance of dry food dog. If not, the unclogged feature will start and dispense the food.
Source	(Apostol, Madroñal, Bucol)
Fit Criterion	The servo motor should dispense the set portion within a ±5% accuracy margin of the target weight.

Requirement No.	5
Event	RFID module to scan unique tags
Description	The RFID module will scan and identify dogs using unique RFID tags assigned to them. When a valid tag is detected, the system will allow feeding to proceed, ensuring that the correct dog is recognized before dispensing food.
Rationale	Integrating RFID ensures individualized feeding by linking food portions to specific dogs. This prevents mix-ups when multiple pets are present and supports the system's goal of providing personalized and fair feeding routines.
Source	(Apostol, Madroñal, Bucol)
Fit Criterion	The RFID module should accurately detect and validate tags within its operating range (3–5 cm) at least 95% of the time.

Functional Requirements

The PAWFEEDS system was designed to ensure reliability, accuracy, and user convenience in automated dog feeding. Its functional requirements cover both software and hardware aspects to meet user expectations and deliver a seamless experience.

Software Functionalities

- **Multi-Pet Profile Management** – Users can switch between multiple dogs (Dog 1, Dog 2) to assign individual feeding profiles. Input birthday (months), weight (kg), food energy density (kcal/100g), sex/neuter status, and activity level.

- **Automated Feeding Calculation** – The system calculates the required food portion (in grams) based on the input data and displays it on the interface.
- **Editable Portion Control** – Users can manually adjust and edit the calculated food portion (grams) before dispensing.
- **Real-Time Notifications** – Alerts are provided for successful feedings, low food storage, skip meal, scheduled meal, ‘clogged system, RFID tags scan for 3x times.
- **Manual Feeding Option** – “Feed Now” button allows immediate dispensing outside scheduled times.

Prototype Development

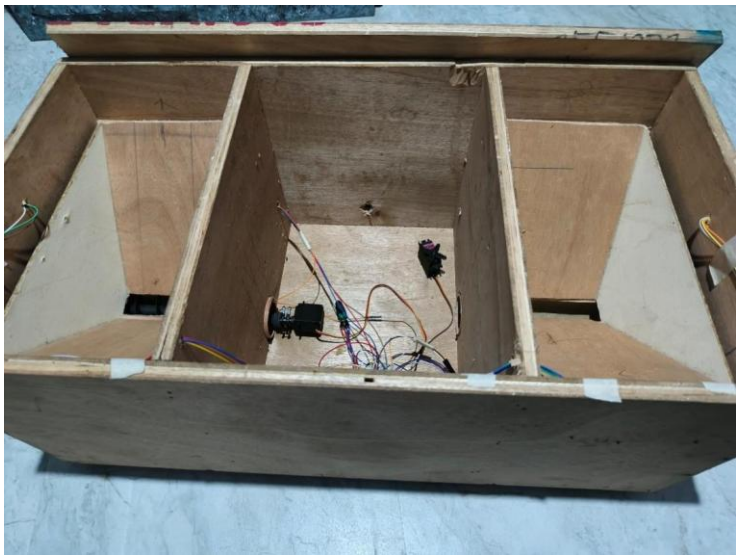
The prototype is an automated dog feed dispenser that is design for household pets and automatically feed dogs depending on the portion size of weight, size, kcal, individual recognition of RFID unique tags, and real-time monitoring. At the center of the system is the ESP32-devkit (master) that function

as the main controller, managing the auger mechanism that is powered by a servo motor.

The supplementary feature which is the RFID module recognition will be added to recognized dogs using assigned RFID unique tags, ensuring that dog profiling and feeding portions are being distributed correctly by the system. Configuring schedules, feeding portions, and real-time monitoring are to be shown in the React Native Mobile application which will allow owners to control the system.

The design of the hardware will consist of having 2 holes that will simultaneously dispense dog foods if an RFID unique tags or a feeding schedule

is being executed. Each hole will be assigned with ESP32-CAM (slave) which will display a stream video to the React Native mobile application.



The inside of the hardware has a food container shaped like a cone, in the middle there is a PVC pipe that the dry food will fall down to that pipe and instantly pushes out by an auger mechanism powered by a servo motor. Below the container is the Weight sensor or the load cell, which calculates the weight of the food in the container and will also calculate the weights of foods being dispense. In the middle of the box is where all the controller and the breadboard of the hardware will be placed, including the wirings of weight sensors, ESP32-CAMs, servo motors, and lastly the power supply.

Hardware Components

Control, Camera, and Processing Components:

ESP-32CAM (slave) – this component acts as the secondary controller. This captures live video streaming allowing users to monitor the live feeding or dispense in real time through the mobile app.

ESP32-CAM Devkit (Master) – the component will act as the Master controller, this will handle the scheduling, process user inputs and coordinate overall with the operation of the feeder.

Dispensing Components:

MG996 Servo Motor – This component gives the needed torque to rotate the auger mechanism.

Auger Mechanism – pushes the dry food and dispenses it in calculated portion sizes based on the user inputs.

Feeding Container – stores the dry food and keeps it clean before dispensing.

Measuring Components:

HX711 Load Cell Amplifier – the load cell is the one that will measure the food container

Identification Components:

RFID Module (ISO 15693) - RFID is used to recognize individual dogs as they approach the feeder, enabling the system to apply the correct feeding schedule and portion according to each dog's stored profile.

RFID Tags – unique identifiers worn by each dog that triggers the system to dispense food after recognizing.

Network and Power Components:

Wi-Fi – this connects the ESP32 devices to the network for app communication.

Power Supply - Provides stable electrical power connections to all electronic components.

Software Components

System Logic

The system logic is developed in C++, which handles the control commands and operations of both the ESP32 Devkit and ESP32-CAM microcontrollers. This layer manages the hardware-level functions, including servo motor activation, RFID tag detection, and live video streaming. It also processes communication with Firebase, which serves as the system's cloud-based data handler for storing schedules, feeding logs, and device status updates.

Application Layer

The mobile application is developed using React Native, serving as the main user interface that allows dog owners to interact with the system. Through the app, users can set and modify feeding schedules, adjust portion sizes, initiate manual feeding, receive real-time notifications, and view live video monitoring. The integration of React Native ensures smooth performance across Android platforms

while maintaining seamless communication with Firebase and the ESP32 controllers for reliable and responsive feeding automation.

Integration with Software

Think of it as a coordinated team where each part has a specific job.
At the heart of the operation are two micro controllers, the brains of the feeder. The first, an ESP32-CAM (slave) that acts as the system's eyes. Its main job is to capture a live video feed of the food bowl and stream it over to your home Wi-Fi. This will allow you to check in at any time using the mobile app to see if your pet has been fed or if the food has been dispensed successfully. This also works with the RFID module, which scans the unique tags of the dog/s, after scanning the food will be dispensed in their own input of how many portions of that pet has to eat.

The second microcontroller, a standard ESP32 (master), which is the master controller that handles all the logic. It's responsible for knowing the feeding schedule that is save on the mobile app, controlling the MG996R servo motor for dispensing the food, calculating the portions sizes base on the inputs of the user, and controlling the auger mechanism through the servo motor. The device ensures

that correct portions are released based on the dog's save inputs in the mobile app.

You, the pet owner will interact with the whole system itself through a REACT native mobile app on your smartphone. The app lets you set time and date of feeding schedules, adjust the portion size, and press the "Feed" button for manual feeding or start interacting. It also displays real time- notifications and streams live video feed that is being captured by our ESP32-CAM (slave).

Finally, the complete flow will look like this, you as the user that uses the device that your smartphone and device are connected to the same network will send a command to over to the Firebase Realtime Database to the ESP32 (master). The ESP32 (master) checks for valid RFID unique tags, and calculates the right inputs and portion and then activates the servo motor to release the food. At the same time, the ESP32-CAM (slave) continues to stream a live feed video so you can open the app and watch the feeding process or dispense process in real-time. Notifications and feeding records are updated instantly in the app, this gives you peace of mind that everything is working out and that your pet has been fed at the right time.

Developmental Tools

The developmental tools used in the PAWFEEDS system consist of hardware components, electronic modules, and supporting devices essential for building and testing the prototype. These tools were selected based on their compatibility, reliability, and role in ensuring that the system functions effectively. The table below presents the list of tools, their specific purposes, unit prices, quantities, and the corresponding total costs.

Table 3: Developmental Tools and Cost

Name	Purpose	Price	Quantity	Total
ESP32Devkit	**Main microcontroller used to control the entire feeder system and handle IoT connectivity.**	**350**	**1**	
ESP32-CAMERA	**Provides live video streaming and image capture for real-time monitoring of the feeding process.**	**600**	**3**	
MG996R	**High-torque motor used to drive the auger mechanism for food dispensing.**	**715**	**4**	
HX711	**Detects and monitors the weight of the food container to determine whether it is empty or has sufficient food.**	**121**	**1**	
RFID/ISO15693	**Detects and identifies individual pets through RFID tags for personalized feeding.**	**3,300**	**1**	
BreadBoard + jumperwires	**Used for prototyping and establishing connections between components.**	**1171**	**1**	
XL4016	**Regulates and steps down voltage to safely power the microcontroller and modules.**	**240**	**1**	
			Grand Total	**9,842**

Testing and Evaluation

To completely validate the “PAWFEEDS” system, our methodology uses a two-pronged evaluation process, beginning with internal functionality testing before proceeding to external user acceptance trials. Originally, our research team will conduct extensive internal checks to assure all technical components are functioning as they are design. This will involve verifying the precision of the food dispenser’s portion control, stable connectivity of the IoT for commands from the android application, and confirming the reliability of the application’s features. At the same time, the live-monitoring video stream will be evaluated for clarity and latency, and the RFID reader’s accuracy on identifying the unique RFID tags. Once the system internal is established, the study will progress to the user acceptance phase in Zamboanga City. We will utilize the purposive sampling numbers of dog owners at least 3 respondents who are actively manages their pets feeding schedule, use an android smartphone, and who voluntarily participate.

The data will be gathered using a researcher made-questionnaire which will get the participants profiles, quantifies the system usability using the 5-point likert scale focusing on usability, functionality, and reliability and collects suggestions. The procedure on the evaluation will involve obtaining consent, providing a complete demonstration of the prototype, and asking the participants to complete some tasks like setting schedules, manual dispensing, and watching the live

streaming video feed. Following some hands-on interaction with the prototype. Finally, all written feedback will be subjected to a thematic analysis to identify common recommendations and key strengths of the system.

CHAPTER IV

RESULTS AND DISCUSSION

Introduction

This chapter presents the complete analysis of data gathered from the testing, technical performance evaluation and user acceptance testing of the PAW FEEDS: An IoT-Based Automated Dog Food Dispenser with an Android application for scheduling and Real-time Monitoring. The findings and testing are all organized and separated into two main sections. First, the technical performance of the hardware components like the ESP32-cam (slave), Load Cell HX711, RFID, RFID tags, Servo Motor, Auger Mechanism, Power Supply, ESP32 (Master), including dispensing accuracy and RFID detection rates. The second is the functionality and usability evaluation of the integrated system and the mobile application

Integration Testing and System Functionality

To validate the PAW FEEDS system readiness, an accurate testing was conducted. This phase ensures that every logic, detection, latency, to the final dispensing are being function according to the system design. The results of these rigorous testing are shown below.

User Account and Provisioning

- **Authentication:** the firebase authentication protocol successfully handled the user authentication from creating an account to login in with google, login sessions without errors. This ensures the secure access to the dashboard.
- **Provisions:** after successfully creating an account the provisioning will follow to set up the PAW FEEDS setup and connecting it to the mobile, ensuring that the mobile application is connected to the mobile and also the feeder system.

Scheduling Module

- **Scheduling Logic:** Users were able to setup and set a specific feeding date and time via android application. The testing confirmed that these schedules were successfully set up and has been sync with the ESP32 (master) internal clock (NTP server sync), ensuring that the physical device

and the mobile application is timely aligned. It can also be customized to add desired grams to the daily portion.

Notification System and logic loops

Testing verified with three alert scenarios below:

- **Scenario A (Scheduled Meal):** The user sets an exact time to scan the tag, the system sends a notification to the mobile application notifying about the feeder ready to dispense, Once the RFID tag has been read it will automatically dispense the set portion of the users and prompting a notification that a tag has been scan.
- **Scenario B (Skip Meal 2 minutes Timeout Protocol):** the feature was tested by allowing a scheduled mealtime to pass without scanning a tag. The internal timer tracks the duration consistently. Exactly 120 seconds (2 minutes) after the scheduled time, the system logic automatically identifies the timeout and triggering the Skip meal notification sending it to the mobile application.
- **Scenario C (Low Food Level):** the load cell continuously monitors the weight of the container, when the container weight dropped to the pre-defined threshold (calibrated during the setup) the system sends a pushed "Low Food Level" warning to the user acting as a proactive inventory management tool.

- **Scenario D (Clogged please check):** The system has an unclogged system to unclog the dog food that is stuck to the auger mechanism.
- **Scenario E (3x RFID tag has been scan):** In this scenario the RFID tag has been scanned more than once, this means that the dog is approaching the system to find food and might be hungry. We added a feature were in after it scanned the RFID tag more than once it will notify the dog owner that their dog might be hungry and therefore can use the manual feeding.

Hardware Execution and Dispensing

- **RFID Authentication:** the testing of RFID both registered and unregistered was successful, registered tags successfully triggers the dispensing servo motor. Un-registered tags have constantly been declined by the system with no food dispensed, this validates the multi-pet identification feature. The dispense food will only happen if the assigned tag has been scanned to the right RFID.
- **Dispensing Mechanism:** With the right validation, the servo motor/auger dispense food with the precise portion calculated by the load cell feedback loop, dispensing the target amount of kibble. This works with a scheduled meal and automatic feed (Feed Now button).

Real-Time Monitoring and Logs

- **Video Streaming:** The ESP32-CAM module successfully displays a transmitted live video to the application. Both camera's relays were

successful thus making the camera on the application smooth and can show the live feed.

- **History Logs:** pets, bowl number, Events like scheduled meals and manual feeding are successfully recorded in the Firebase Realtime database and displayed in the app's History tab for user to view or to be downloaded as an invoice.

Component/Feature	Text Case Desc.	Expected Outcome	Actual Result	Status
User Account	**User registration**	**User Accesses the dashboard securely.**	**Authentication successful via Firebase.**	**Passed**
Provisioning	**Provision/setup**	**User accesses the dashboard without errors to login.**	**Setup successful mobile and feeder recorded via Firebase.**	**Passed**
Scheduling	**Set/Edit Feeding time.**	**App saves time to the database.**	**Scheduled sync to ESP32**	**Passed**
Notification (Scheduled meal)	**Schedule meal alert**	**Pop-up when a scheduled meal matches real time.**	**Alert received; system enters "scan mode"**	**Passed**
Notification (Low food level/ clogged system)	**Low Food Level alert**	**Pop-up when the load cell doesn't sense any weight from the bowl or the dog food has been clogged**	**Alert received. System notifies the user "Low food level or the clogged auger"**	**Passed**

Component/Feature	Text Case Desc.	Expected Outcome	Actual Result	Status
RFID Authentication	Scan Registered Tag.	Recognize specific pet ID.	ID matched; dispensing triggered.	Passed
RFID Rejection	Scan Unregistered tag	Deny access to food dispense.	No food dispensed into bowl.	Passed
Dispensing Logic	Servo Motor activation	Auger rotates and stop until the target weight of the bowl has been achieved.	Food dispensed into bowl.	Passed
Skip meal logic	No tag detected for 120 seconds (2 minutes)	Timer expires after 120s.	"Skip Meal" notification triggered.	Passed
Manual Skip	User clicks only the turn on or off button from the scheduled meals.	The scheduled meal is skip and calculations are adjusted.	Database updated status recorded.	Passed
Video stream	Access camera feed	Live video feed display on the app	Video visible (quality varies by Internet or Power)	Passed
History Logs	View Feeding Records (pets, bowl number, schedule meals and manual feeding)	Show time, date, grams dispensed and status.	Accurate history displayed in App.	Passed

Technical Performance and Latency Analysis

Beyond verifying that the features simply “worked” (binary functionality) the system’s performance was assessed to measure its reliability, speed, and accuracy under varying real-world conditions. This particular section points the responsiveness of the notification ecosystem and the physical precision of the hardware components.

Network Dependency and Latency Evaluation

A finding during the testing phase is that the system is reliant on the internet connectivity. While the internal logic of the ESP32 such as the 2-minute timer for the skip meals, functioned precisely on the device, the transmission of data to the user’s phone experienced measurable delays.

- **Notification Latency**: Testing revealed that “skip meal”, clogged or empty container, 3x times RFID tag has been scan, and “scan tag successfully” notifications arrived with a latency of 10-15 seconds. However, the delays were consistent across different network speeds, assigned to the processing time required for the ESP32 to communicate with the firebase Realtime Database and for the Android app to vote these changes.

- **Video Stream Stability**: the video monitoring feature was limited by the ESP32-CAM hardware capabilities. The stream is capped at 15 Frames per seconds (FPS). During testing, we observed that the video was functioning very well, but the delay 10-15 seconds meant that the user is viewing the feeding event slightly after the scene had occurred, rather than in real-time.

Hardware Accuracy and Sensitivity Metrics

The Physical components were tested to establish their error and success rates.

- **RFID Detection Rate:** The RFID system was tested across 100 trials. It achieved 95% success rate scans. The remaining failed scans were specifically observed when the tag was move rapidly across the sensor, simulating a dog running past the feeder. The testing confirmed that for successful detection, the tag is now required to momentarily pause or move slower within the 3-5cm range of the reader.

- **Precise Dispense**: The load cell maintained an average of 5-10 grams portioning errors depending on the auger mechanism because of each wing is dispensed by batch and not by the actual weight.

Metric	Target	Actual Result	Remarks
Portion Accuracy	**5-10 grams**	**3-5 grams**	**Variance due to irregular kibble shapes.**
RFID Detection	**100%**	**95%**	**95/100 trials. Failed when scanning too fast.**
Video Frame Rate	**30 FPS**	**15 FPS**	**Hardware limitation of ESP32-CAM.**
System Latency	**5 seconds**	**10-15 seconds**	**Highly Dependent on the internet speed and cloud processing.**

RFID Sensitivity and Speed: the RFID reader achieved a 95% success rate (95 to 100 trials). The 5 failed trials occurred when the tag was scanned at a high speed, simulating a dog running to the dispenser or a dog running by. The feeder required the tag to be momentarily steady within the 3-5cm range to register successfully.

Video Limitations: the video stream was limited to 15 FPS. During testing, the feed appeared choppy and lag when the internet connection was slow or unstable. Additionally, the 10-15 seconds latency meant the owner saw the dog on the live feed slightly late after the event began.

Dispensing Variance: The 5 grams error margin was allocated to the mechanical "Crushing" of irregular kibble shapes during the auger mechanism rotation.

Survey Results

Before we pick who will evaluate the device hands on, we conducted a survey using the google forms to understand the habits and problems of dog owners. We gathered data from 30 respondents the result helped us confirm that these busy owners need an automated feeder like PAWFEEDS. Below are the test results that each respondent answered in the google form survey.

1	Timestamp	Email Address	Name	Do you agree to participate in this survey?	Do you own dog/s?	How many dog/s do you own?	What type of food do you usually feed your dog	how many times a day do you feed your dog/s?	Are your dog/s a picky eater?
2	1/25/2026 19:57:12	maryjasminesalazar@gm	Mary Jasmine B. Salazar	Yes	Yes	more than 2	Wet food	Twice	No
3	1/25/2026 20:01:24	averfellysalazar15@gmai	Averfelly B. Salazar	Yes	Yes	more than 2	Wet food	Twice	No
4	1/25/2026 20:11:09	mselleebuencuchillo@gm	Ella Jade S. Buencuchillo	Yes	Yes	more than 2	Both	Twice	Sometimes
5	1/27/2026 9:38:36	eh202201370@wmsu.ed	Aldwin N. Suarez	Yes	Yes	2	Both	More than twice	No
6	1/27/2026 9:41:45	eh202203221@wmsu.ed	Ashlyn Calaor	Yes	Yes	1	Both	More than twice	No
7	1/27/2026 9:43:09	eh202201761@wmsu.ed	Eubert Atilano	Yes	Yes	1	Both	Twice	No
8	1/27/2026 10:07:44	villarcamponoellenepeark	Noellene Pearl Villarcamp	Yes	Yes	1	Wet food	Twice	No
9	1/27/2026 10:20:49	eh202204032@wmsu.ed	Reigh	Yes	Yes	1	Both	Twice	Sometimes
10	1/27/2026 10:55:15	janchristianarsena@gma	Jan Christian M. Arseña	Yes	Yes	1	Both	Twice	Sometimes
11	1/27/2026 20:52:38	inoviaabigailblanche@gm	Inovia, Abigail Blanche F.	Yes	Yes	more than 2	Both	More than twice	Sometimes
12	1/27/2026 20:56:00	leolabartinos193@gmail.c	leo	Yes	Yes	more than 2	Both	Twice	Sometimes
13	1/27/2026 21:00:02	johnethandapolinario@gi	John Ethan Apolinario	Yes	Yes	1	Both	More than twice	Sometimes
14	1/27/2026 21:02:39	shiezsablin@gmail.com	Shiez Sablin	Yes	Yes	1	Both	More than twice	Sometimes
15	1/27/2026 21:07:39	samorinladyshane@gmai	Samorin, Lady Shane S.	Yes	Yes	1	Both	Twice	No
16	1/27/2026 21:07:59	reignfrye9@gmail.com	PORTACIONI, KYLA REGI	Yes	Yes	1	Dry food	More than twice	No
17	1/27/2026 21:08:23	amisoles03@gmail.com	Angel Misoles	Yes	Yes	2	Dry food	More than twice	No
18	1/27/2026 21:09:16	denebelee02@gmail.co	Dynabelle Gutiol	Yes	Yes	more than 2	Both	Twice	Sometimes
19	1/27/2026 21:11:50	anmumtahanajailani@gm	Jailani An-mumtahana I.	Yes	No	1	Both	Twice	No
20	1/27/2026 21:10:54	marlandreyquiachon@gm	Marl Andrey	Yes	Yes	1	Both	Twice	Sometimes
21	1/27/2026 21:12:54	guzmanjanahrose@gmail	Janah Rose	Yes	Yes	2	Both	More than twice	Sometimes
22	1/27/2026 21:16:47	chanjessa02@gmail.com	Jessa	Yes	Yes	more than 2	Both	Twice	Sometimes
23	1/27/2026 21:18:31	eh202201968@wmsu.ed	TANLIAN ROYD B.	Yes	Yes	more than 2	Both	Twice	Sometimes
24	1/27/2026 21:20:17	marisajadjaie7@gmail.co	Jadjaie Marisa T.	Yes	No	1	Both	Twice	Sometimes
25	1/27/2026 21:30:32	eh202201447@wmsu.ed	John Erick Anacleto	Yes	Yes	1	Both	Once	No
26	1/27/2026 21:36:44	hs202304119@wmsu.ed	Vincent	Yes	Yes	2	Wet food	More than twice	No

27	1/27/2026 21:39:56	hadjulasittiayesha2021@	Ayesha Hadjula	Yes	Yes		1	Both	More than twice	Yes
28	1/27/2026 21:50:58	franckarming2003@gmail	Franck Jumar P. Arning	Yes	Yes	more than 2		Both	More than twice	Yes
29	1/27/2026 21:59:04	jaycelijane123@gmail.cor	Jaycel Jane Sales	Yes	Yes		1	Both	More than twice	Sometimes
30	1/27/2026 22:04:04	naenaebusaing@gmail.c	Vianneye Busaing	Yes	Yes	more than 2		Both	More than twice	No

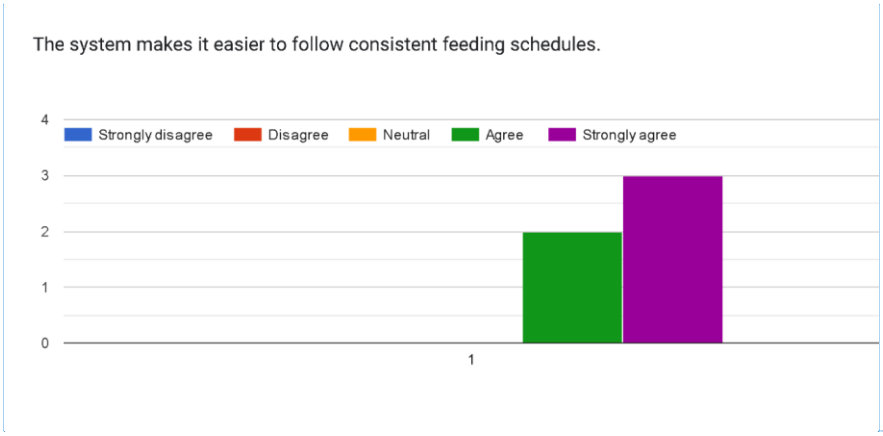
Out of 30 respondents, 28 confirmed that they own a dog and 2 respondents answered that they own dogs. Which means that the results and data comes from real experiences dog owners.

Questions	**Top Answers**	**Frequency (respondents)**
1. Do you own a dog?	Yes	28
2. How many dogs do you own?	1 Dog	15
3. What type of food do you use?	Both (Dry & Wet)	23
4. How often do you feed them?	Twice a day	15
5. Is your dog a picky eater?	Sometimes	15

Half of the owners have only 1 dog. However, the other half owns multiple dogs, which supports our decision to include RFID technology to identify each pet and avoid conflict during feeding times.

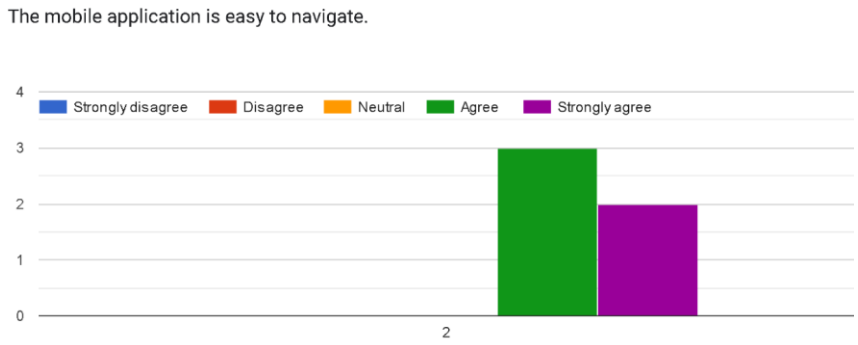
The 23 respondents which is a large majority said that they used both dry and wet food to feed their dogs. But our device only handles dry food which is specifically built for dry kibble. the device allows the owners to change or manage the diets and dry feeds that they put it in the device food storage. The results showed that feeding is a frequent task especially for busy owners. 15 respondents feed their dogs twice a day, while 14 of the respondents feed them more than twice. This shows that owners spend a lot of time feeding their pets, and an automated scheduler would save all the time and effort even if their busy.

When the 15 respondents answered that “Sometimes” their dogs are picky eaters and 13 said “No” since many dogs are picky, the portion control of the PAWFEEDS is important to ensure that they eat right amount and at the right time.



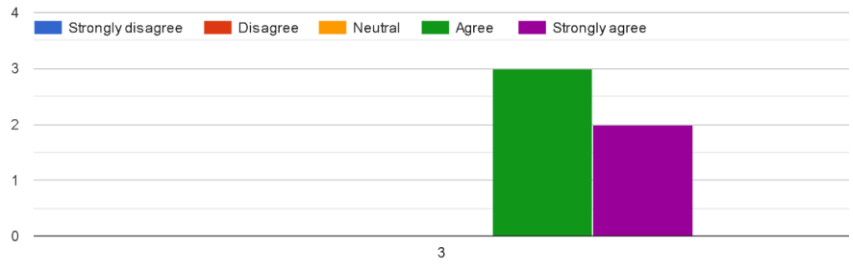
Commented [JA2]: Explain every result in all graphs.

This graph shows that 3 evaluated it as “Strongly Agree” that the system is easier to follow and it maintains a consistent feeding schedules. While, 2 respondents rated it as “Agree”. This shows that the evaluators are satisfied with the system.



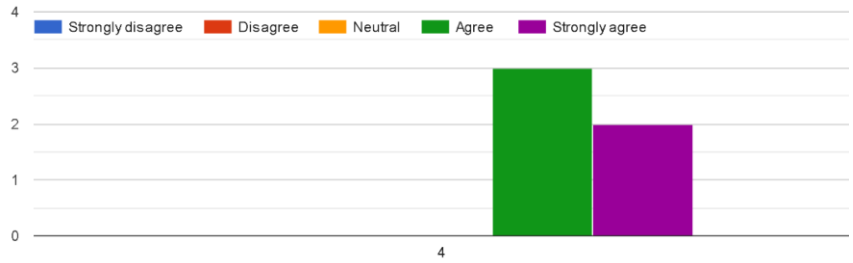
3 respondents evaluated it as “Agree” and 2 for “Strongly Agree”, this graph explains the result of how respondents are well satisfied with the mobile application interface and navigations.

Adjusting input feeding times or portion sizes through the app is convenient.



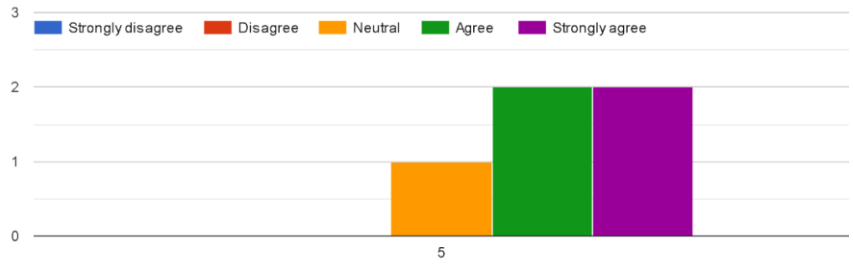
3 respondents evaluated it as “Agree” and 2 for “Strongly Agree”. The graphs shows that the feature of adjusting the feeding times, portion sizes and etc. are convenient and reliable to help them manage their pets’ feeding schedules.

The PAWFEEDS system is good and reliable for dog owners.



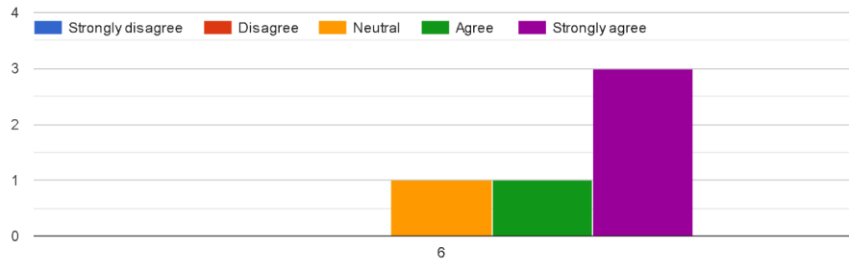
3 respondents evaluated it as “Agree” and 2 for “Strongly Agree”. The graphs shows that the PAWFEEDS system app is good and reliable among dog owners and can be trusted with how they manage their pets in a modern way.

The live streaming video is clear and can easily monitor the dispensed food.



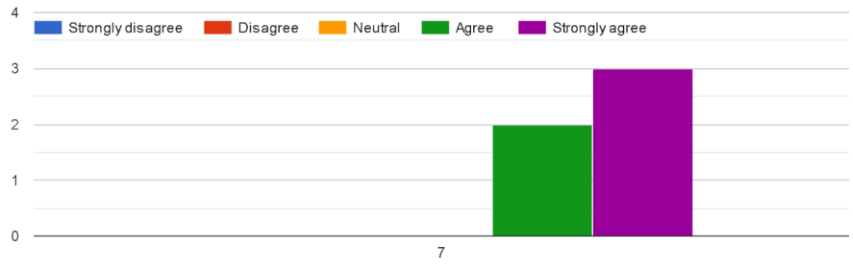
2 respondents evaluated it as “Agree”, 2 evaluated is as “Strongly Agree” and 1 evaluated it as “Neutral”. The graphs shows that 1 evaluated it as Neutral because of the live streaming video is maybe not clear but this is also because of how the internet is strong in their area. The live streaming video and monitoring is highly dependent on how strong is the internet is, with the other 4 respondents being satisfied with the live stream and monitoring in this case they may have a strong internet connection so the live streaming is clear.

The system dispensed food accurate portion sizes based on the dog/s needs.



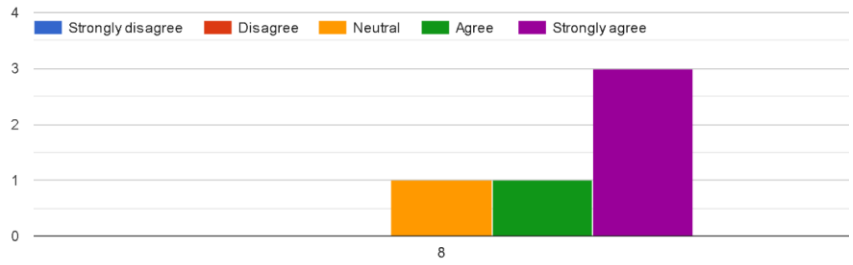
3 respondents evaluated it as “Strongly Agree”, 1 respondent evaluated it as “Agree” and 1 respondent evaluated it as “Neutral”. The graph explains shows that the system dispensed food accurate portion sizes based on the dog/s needs. More than 3 evaluators are satisfied with how accurate is the portion sizes, while 1 evaluated it as neutral. This explains that the system can have an accuracy error of less and more 5 grams dispense. The dispensed food is also highly based of on how small, medium and large is the dry food being dispensed.

Automated Feeding is better than old traditional feeding.



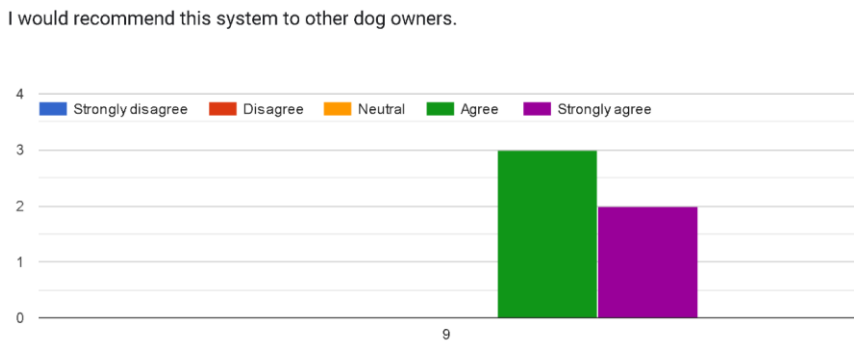
3 respondents evaluated it as “Strongly Agree”, 2 respondents evaluated it as “Agree”. The graph explains that how well is the automated feeding better than the traditional manual feeding. the evaluators are satisfied on the automated feeding because of how less time consuming it is slightly better than the old manual traditional feeding.

The RFID feature is working correctly and recognize correct dog tags.



This graph shows that the RFID feature is working correctly and can recognize correct dog tags. While some respondents are not satisfied well enough of the

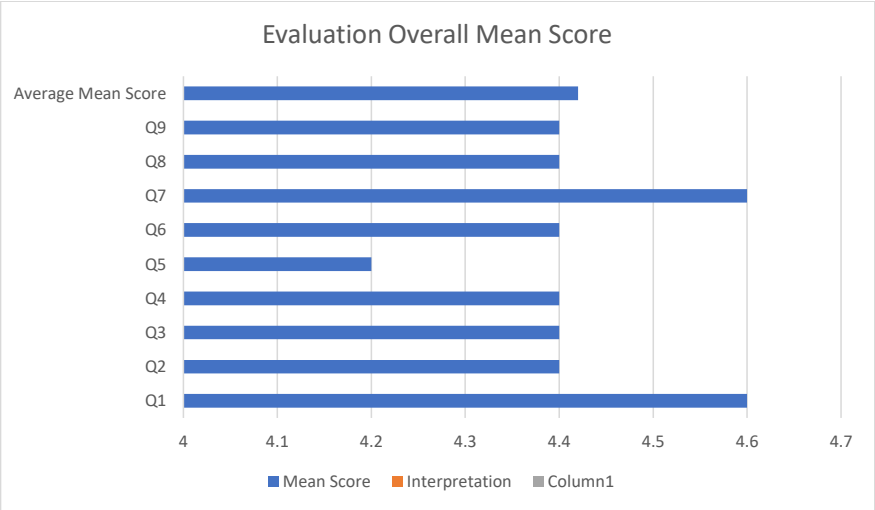
feature, this just shows that the RFID feature is highly dependent on how strong is the internet speed and how fast can it trigger the system. This can be improved by making the internet speed faster so that tags can be easily recognize.



The evaluators expressed high level of satisfaction with the system and would recommend this system to other dog owners. This shows that the system developed effectively addresses the needs of dog owners and use it as a valuable tool within it's target.

Evaluation Results

To see if we met our goals, we had the 5 evaluators test the system hands-on. We used a 5 point-likert scale to rate the specific features of PAWFEEDS. The table present below is the total weighted mean of each evaluation statement.



As shown above in the table, the device received an overall mean score of 4.2 which will fall under the category of “Strongly Agree”. This data confirms that the evaluators found the system highly effective.

- **Hardware Accuracy**: the users gave a score of 4.40 for both the food dispensing and RFID accuracy. While most users rate “Strongly Agree” that the hardware works, one of the evaluators gave a “Neutral” rating. This

explains that while the system is functional and working, the RFID tag needs to properly position and timing to be able to detect immediately.

- **Software Usability**: The Scheduling feature of the app received the highest score of 4.60 proving that the automation is the strongest feature of the system. However, the live feed camera received the lowest mean score averaging of 4.20, due to likely to poor internet connection speeds affecting the live feed video clarity and quality. Despite all of this, the rating still received “Strongly Agree” which means the video is clear enough for use of monitoring.
- **Overall Impact**: Even with these scores the evaluators still agreed with a high score of 4.60 that automated feeding is better than traditional methods. This proves that the system of PAWFEEDS and device has successfully solved the problem of convenience for busy dogs’ owners.

Qualitative Feedback (Open Ended Responses)

PART B: OPEN ENDED QUESTIONS

what do you like mostly about our system?

5 responses

- The remote feeding control via smartphone.
- What I like the most about it is its scheduled feeding function which it provides swift access and convenience especially in having exact feeding time and offering a healthy mechanism for dogs in terms of dietary intake. It is also user-friendly and efficient.
- easy to navigate, and offers user friendly interface.
- Easy to use, user-friendly
- The scheduled feeding and RFID is very convenient for dog identification and proper meal distribution

What features would you suggest on improving or adding?

5 responses

- None
- The quality of live camera feature
- none.
- Nothing

In Part C, of the evaluation, we asked the 5 evaluators to write down any specific comments or suggestions that they had in mind while testing out hands-on of the device of PAWFEEDS system. This qualitative data helps explain the scores we got in previous section.

In the open-ended questions, one of the respondents commented about what they liked about the system which targets one of our objectives which is the RFID functionality. The respondents commented “the scheduled feeding and RFID is very convenient for dog identification and proper meal distribution”. Which means that the systems RFID detection is working and functioning but it is still base on how fast and how far away the trigger of the detection the dog did.

One of the evaluators also mentions about the quality of the camera that needs to be improve. While the evaluators appreciated the ability to monitor their dogs, 1 out of 5 mentioned that the current resolution is sometimes laggy or maybe not moving at all.

Verification of Objectives

To conclude the results and discussion, the researchers mapped the findings from 30 survey respondents, 5 evaluators against the specific objectives of this study. The verification confirms that the system successfully met its goals.

Objective 1: Design a food dispensing mechanism for dogs. (ACHIEVED)

- The auger-based mechanism was successfully designed and tested. Question 6 receiving a high score of 4.40 mean score (Strongly Agree)

Objective 2: Create a React Android App for scheduling and alerts (ACHIEVED)

- The mobile application created by React was successfully developed and deployed. The evaluation showed that schedule consistency (question 1) received the highest rating of 4.60 and App navigation (question 2) received a 4.40 proving the app is functional and user-friendly.

Objective 3: Assess system accuracy (hardware) and usability (user acceptance). (ACHIEVED)

- **Hardware accuracy:** Confirmed by technical test showing that 95% RFID detection rate and a maximum of dispensing errors only 5 grams.
- **Usability:** Confirmed by the 5 evaluators who gave the system an Overall mean score of 4.42 interpreted as “Strongly Agree”.

Objective 4: Construct a dual-dispensing mechanism with auger control.

- The dispense function correctly during testing. The auger control successfully manages to dispense the portions, validated by the 4.40 rating for portion control and accuracy to handle dry kibble.

Objective 5: Incorporate RFID-based recognition as a supporting feature to identify multiple dogs using separate unique tags.

- The features address the need of the 15 respondents who owns multiple dogs. The RFID system achieved of 95% success rate in technical trials,

and evaluators tag recognition at 4.40 proving it efficient and prevent food stealing.

Objective 6: Compare the system processes/performances to the traditional feeding.

- We asked the evaluators to compare the two methods, the tradition and the automated feeding. The score that automated feeding is better than traditional feeding is roughly 4.60 interpreted as “Strongly Agree” confirming that the system offers superior efficiency compared to manual scoops and bowls.

Objective 7: Evaluate the system usability among the pet owners.

- The system was tested by 5 evaluators hands-on. Their feedback regarding the video quality (question 5), and overall reliability (Question 4) provided the necessary data to conclude that the system is highly usable.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Conclusion

This study aimed to design the PAW FEEDS: an IoT-Based Automated Dog Food Dispenser with an Android Application for Scheduling and Real-Time Monitoring. Based on the data gathered from the technical performance tests and the User Acceptance testing (UAT) with five dog owners, the researchers arrived at the following conclusions:

Technical Viability and Accuracy: The system successfully met all the core functional requirements. The load cell integration for precise diet management with a dispensing error of 5 grams max. the RFID authentication system proved that unreliable for multi-pet environments, successfully achieve a 95% detection rate (95 out of 100 trials) provided the tag was within the range.

Logic and Functionality: the system logic was validated as robust. “Low food level” sensor effectively prevented empty dispensing and the “skip meal” protocol was a success identifying missed feedings after the 2 minutes timeout.

User Acceptance: the system received a high acceptability rate, confirming that dog owners valued the ability to schedule feeds remotely and the peace of mind provided by the notifications.

However, the study concludes that while the hardware automation is reliable, the real-time monitoring is strictly reliant to the hardware and internet limitations.

Explicit Limitations: Despite its successful implementation, the prototype exhibited specific difficulties during testing phase:

- The ESP32-CAM module is limited the video stream to a maximum of 15 FPS. Additionally, network process has been resulted to 10-15 seconds of delay between the actual event and the user's view on the app.
- While accurate scanning is high, the RFID reader requires the pet to move slow or pause within the range of the reader
- The "skip meal", "low food level" and "scan successfully" notifications are highly dependent on the network. Connection lost results to lost or delay of pop ups.
- The auger mechanism is design only for kibble or dry food. Thus, it cannot dispense wet, moist, large and irregular treats.

Recommendations

To address these limitations and prepare the system for a potential commercial experience, the following improvements are prioritized:

- **Hardware Upgrade for Real-Time Monitoring:** Future iterations should replace the ESP-32CAM with a dedicated IP Camera module or a higher performance micro controller to support the 30FPS live streaming and reduce the latency of the camera to 5 seconds.
- **Offline Data Logging:** Integrate a local storage module (SD card) to record the feeding logs locally.
- **Enhance RFID Range:** Upgrade to a UHF (Ultra High Frequency) RFID Reader. This would extend the scanning range more and it will ensure that even when the dog is fast it can still dispense.

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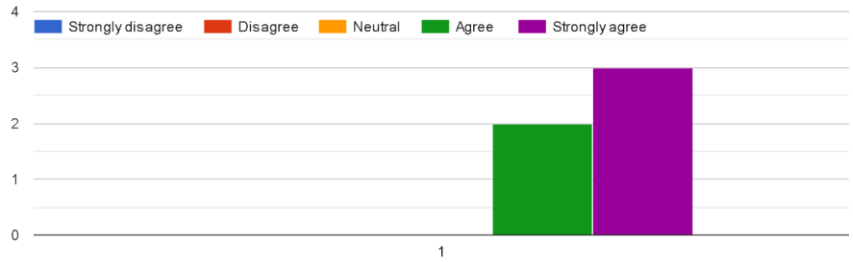
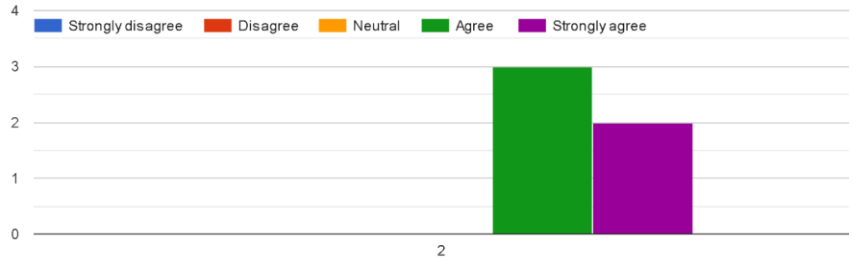
Appendix A: Data Collected

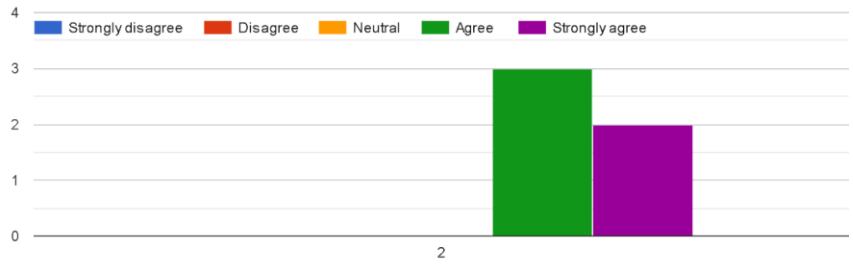
The system makes it easier to follow consistent feeding schedules.



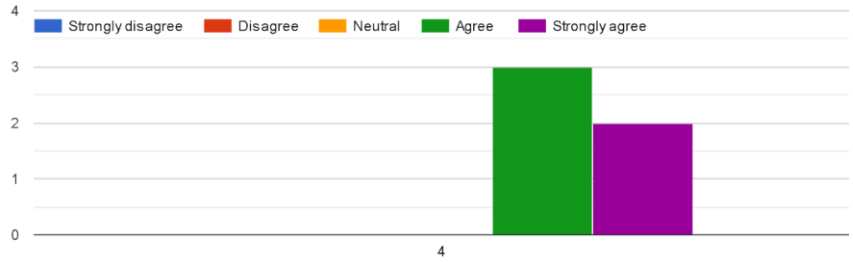
The mobile application is easy to navigate.



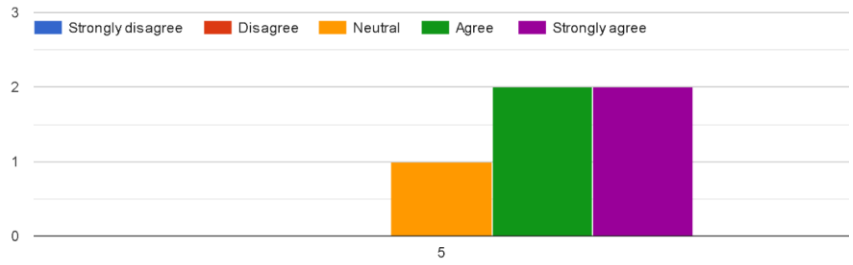
The mobile application is easy to navigate.



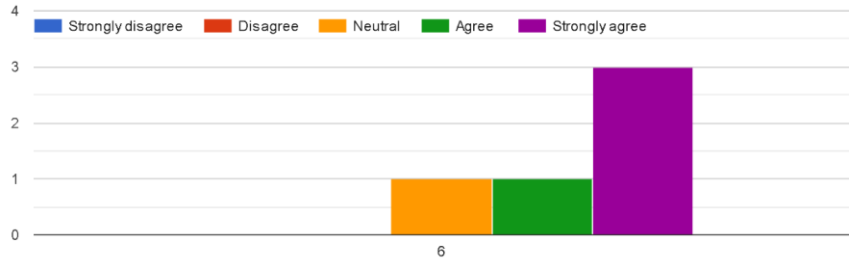
The PAWFEEDS system is good and reliable for dog owners.



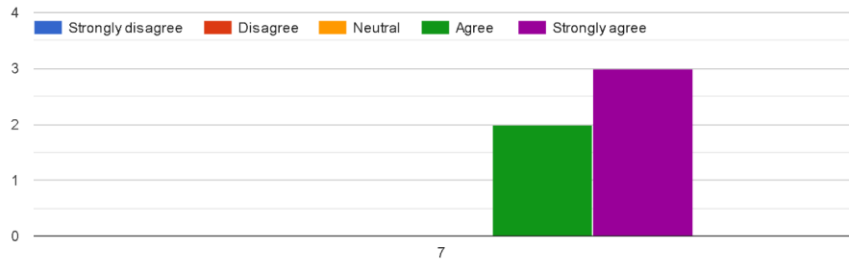
The live streaming video is clear and can easily monitor the dispensed food.



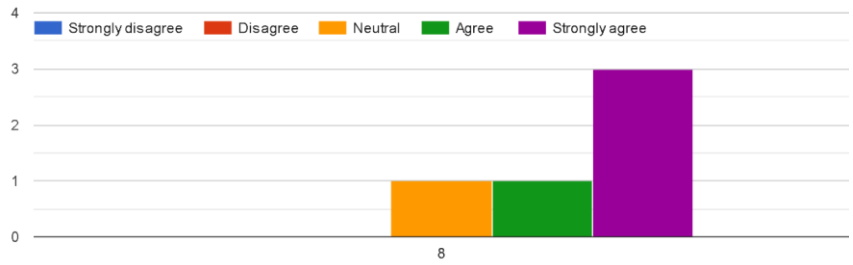
The system dispensed food accurate portion sizes based on the dog/s needs.



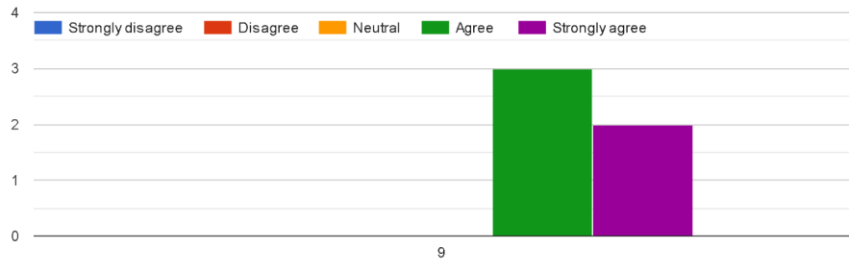
Automated Feeding is better than old traditional feeding.

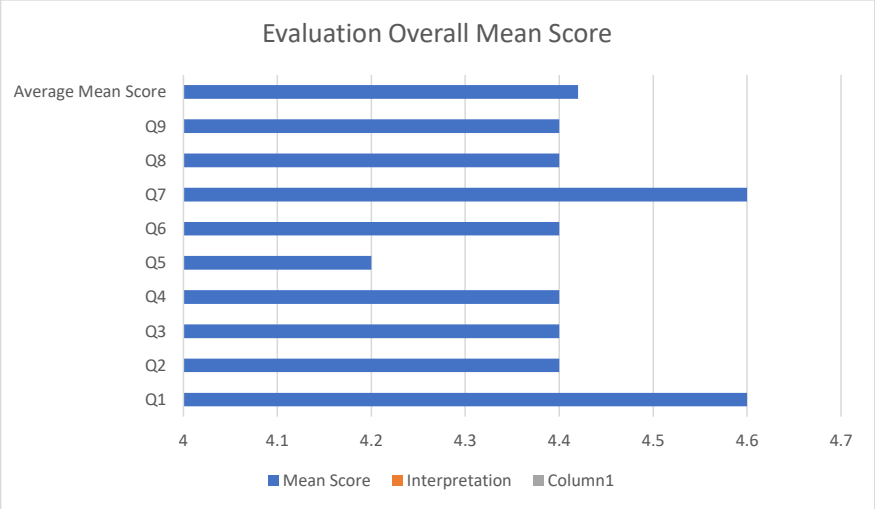


The RFID feature is working correctly and recognize correct dog tags.



I would recommend this system to other dog owners.





PART B: OPEN ENDED QUESTIONS

what do you like mostly about our system?

5 responses

The remote feeding control via smartphone.

What I like the most about it is its scheduled feeding function which it provides swift access and convenience especially in having exact feeding time and offering a healthy mechanism for dogs in terms of dietary intake. It is also user-friendly and efficient.

easy to navigate, and offers user friendly interface.

Easy to use, user-friendly

The scheduled feeding and RFID is very convenient for dog identification and proper meal distribution

What features would you suggest on improving or adding?

5 responses

None

The quality of live camera feature

none.

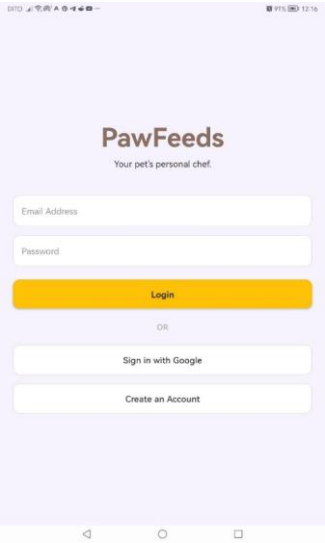
Nothing

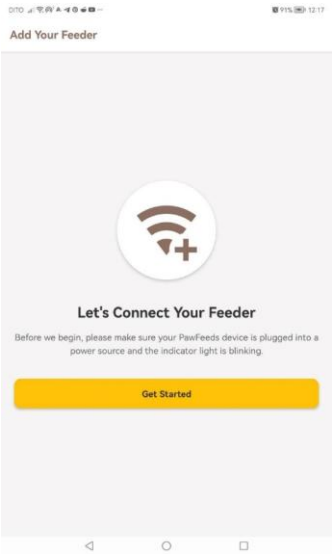
1	Timestamp	Email Address	Name	Do you agree to participate in this survey?	Do you own dog/s?	How many dog/s do you own?	What type of food do you usually feed your dog	how many times a day do you feed your dog/s?	Are your dog/s a picky eater?
2	1/25/2026 19:57:12	maryjasminesalazar@gr	Mary Jasmine B. Salazar	Yes	Yes	more than 2	Wet food	Twice	No
3	1/25/2026 20:01:24	averfellysalazar15@gmai	Averfelly B. Salazar	Yes	Yes	more than 2	Wet food	Twice	No
4	1/25/2026 20:11:09	msellebuencuchillo@gm	Ella Jade S. Buencuchillo	Yes	Yes	more than 2	Both	Twice	Sometimes
5	1/27/2026 9:38:36	eh202201370@wmsu.edu	Aldwin N. Suarez	Yes	Yes	2	Both	More than twice	No
6	1/27/2026 9:41:45	eh202203221@wmsu.edu	Ashlyn Calaor	Yes	Yes	1	Both	More than twice	No
7	1/27/2026 9:43:09	eh202201761@wmsu.edu	Eubert Atiliano	Yes	Yes	1	Both	Twice	No
8	1/27/2026 10:07:44	villarcampnoellenepearl	Noellene Pearl Villarcamp	Yes	Yes	1	Wet food	Twice	No
9	1/27/2026 10:20:49	eh202204032@wmsu.edu	Reigh	Yes	Yes	1	Both	Twice	Sometimes
10	1/27/2026 10:55:15	janchristanarsena@gma	Jan Christian M. Arseña	Yes	Yes	1	Both	Twice	Sometimes
11	1/27/2026 20:52:38	inovaabigailbianche@gr	Inovia, Abigail Bianche F.	Yes	Yes	more than 2	Both	More than twice	Sometimes
12	1/27/2026 20:56:00	leolabartinos193@gmail.c	leo	Yes	Yes	more than 2	Both	Twice	Sometimes
13	1/27/2026 21:00:02	johnethandapolinario@gr	John Ethan Apolinario	Yes	Yes	1	Both	More than twice	Sometimes
14	1/27/2026 21:02:39	shiezsablin@gmail.com	Shiez Sablin	Yes	Yes	1	Both	More than twice	Sometimes
15	1/27/2026 21:07:39	samorinladyshane@gmai	Samorin, Lady Shane S.	Yes	Yes	1	Both	Twice	No
16	1/27/2026 21:07:59	reignfrye9@gmail.com	PORTACION, KYLA REIGH	Yes	Yes	1	Dry food	More than twice	No
17	1/27/2026 21:08:23	amisoles03@gmail.com	Angel Misoles	Yes	Yes	2	Dry food	More than twice	No
18	1/27/2026 21:09:16	denebelee028@gmail.co	Dynabelle Quitiol	Yes	Yes	more than 2	Both	Twice	Sometimes
19	1/27/2026 21:11:50	annumtahanajalani@gr	Jalani An-muntahana I.	Yes	No	1	Both	Twice	No
20	1/27/2026 21:18:54	marlandreyquiachon@gr	Marl Andrey	Yes	Yes	1	Both	Twice	Sometimes
21	1/27/2026 21:12:54	guzmanjanahrose@gmail	Janah Rose	Yes	Yes	2	Both	More than twice	Sometimes
22	1/27/2026 21:16:47	chanjessa22@gmail.com	Jessa	Yes	Yes	more than 2	Both	Twice	Sometimes
23	1/27/2026 21:18:31	eh202201966@wmsu.edu	TAN, IAN ROYD B.	Yes	Yes	more than 2	Both	Twice	Sometimes
24	1/27/2026 21:20:17	marisajadjale7@gmail.co	Jadjale Marisa T.	Yes	No	1	Both	Twice	Sometimes
25	1/27/2026 21:30:32	eh202201447@wmsu.edu	John Erick Anacleto	Yes	Yes	1	Both	Once	No
26	1/27/2026 21:36:44	hz202204119@wmsu.edu	Vincent	Yes	Yes	2	Wet food	More than twice	No
27	1/27/2026 21:38:56	hadjulaashtayesha2021@	Ayesha Hadjula	Yes	Yes	1	Both	More than twice	Yes
28	1/27/2026 21:50:58	franckarning2003@gmail	Franck Jurnar P. Arning	Yes	Yes	more than 2	Both	More than twice	Yes
29	1/27/2026 21:59:04	jaycelijane123@gmail.cor	Jaycel Jane Sales	Yes	Yes	1	Both	More than twice	Sometimes
30	1/27/2026 22:04:04	naenaebusaing@gmail.c	Vianneye Busaing	Yes	Yes	more than 2	Both	More than twice	No

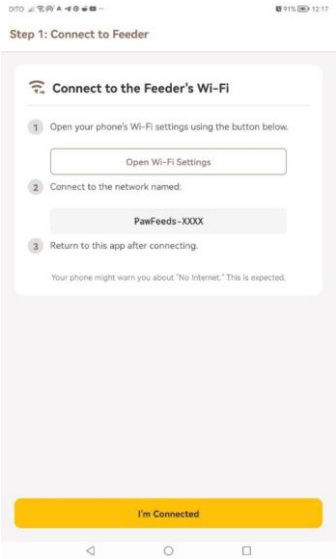
Commented [JA3]: What is the chart al about? Provide explanation

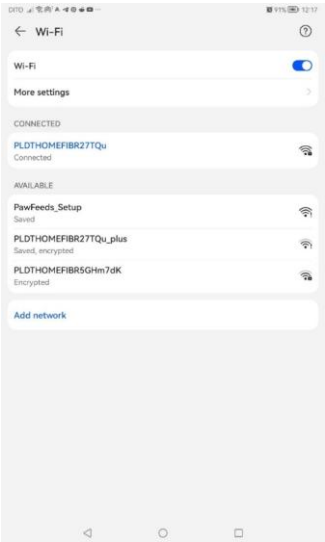
Appendix B: System Design

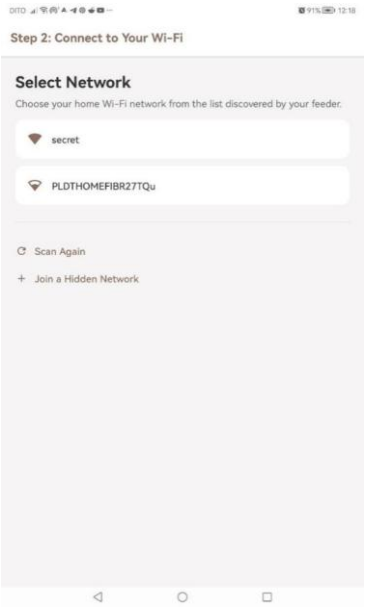
Note: include user interface, system architecture or system flowchart/diagram.

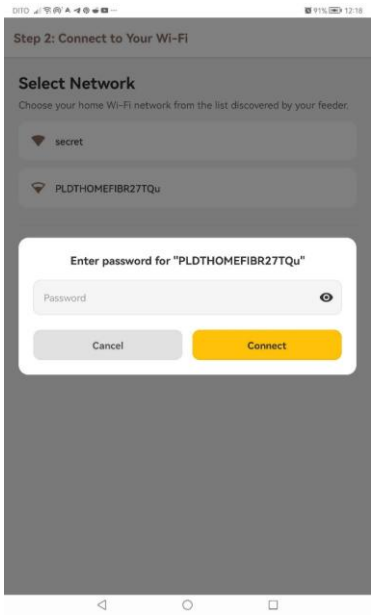


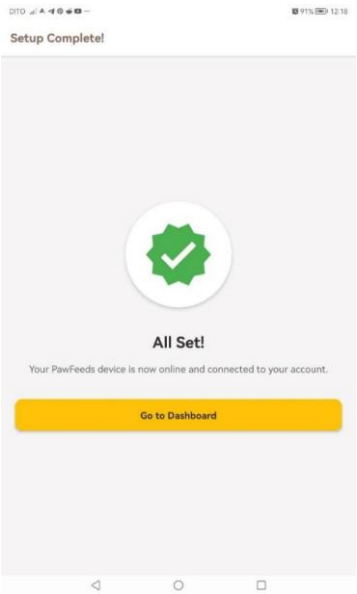


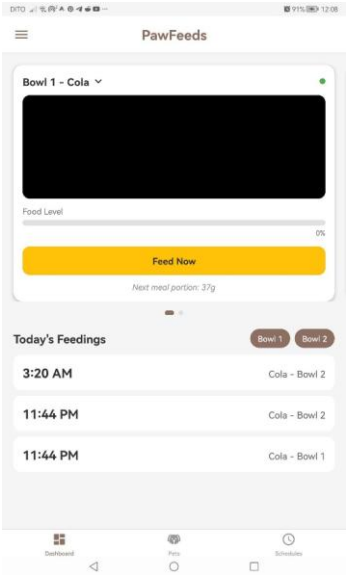


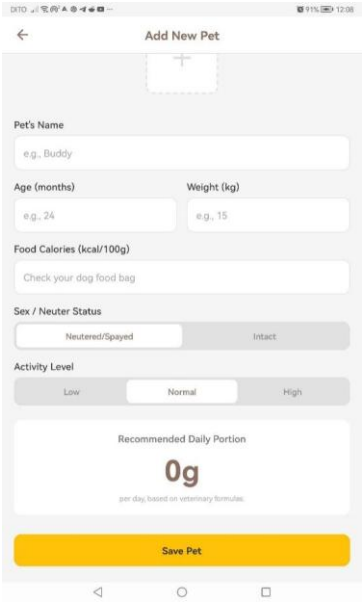


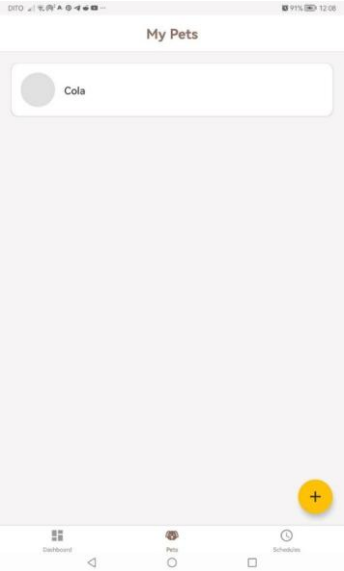

Step 2: Connect to Your Wi-Fi
Select Network
Choose your home Wi-Fi network from the list discovered by your feeder.
secret
PLDTHOMEFIBR27TQu
Scan Again
Join a Hidden Network

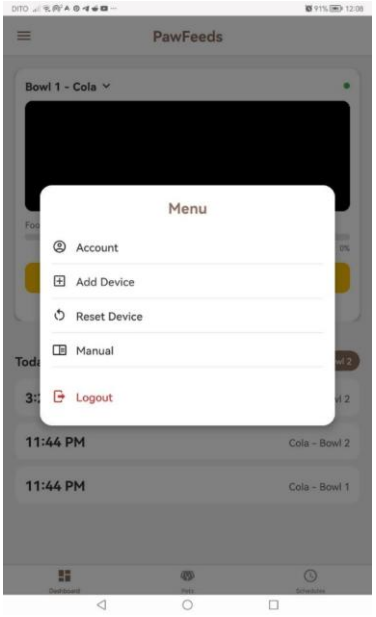

Step 2: Connect to Your Wi-Fi
Select Network
Choose your home Wi-Fi network from the list discovered by your feeder.
secret
PLDTHOMEFIBR27TQu
Enter password for "PLDTHOMEFIBR27TQu"
Password
Cancel
Connect

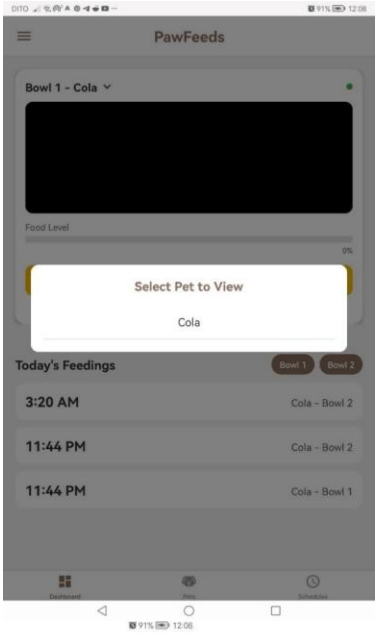

Setup Complete!
All Set!
Your PawFeeds device is now online and connected to your account.
Go to Dashboard

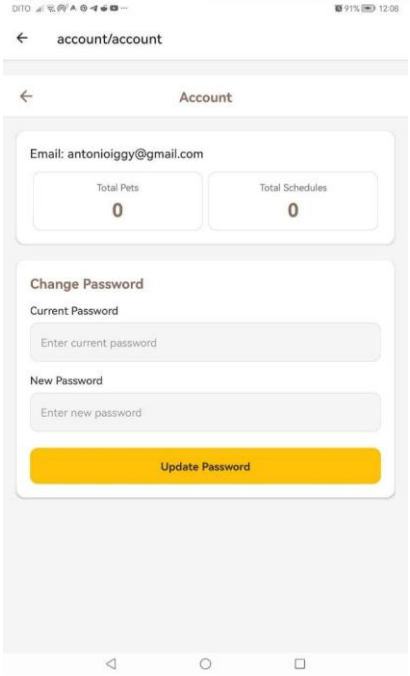

PawFeeds
Bowl 1 - Cola
Food Level
0%
Feed Now
Next meal portion: 37g
Today's Feedings
Bowl 1
Bowl 2
3:20 AM
Cola - Bowl 2
11:44 PM
Cola - Bowl 2
11:44 PM
Cola - Bowl 1

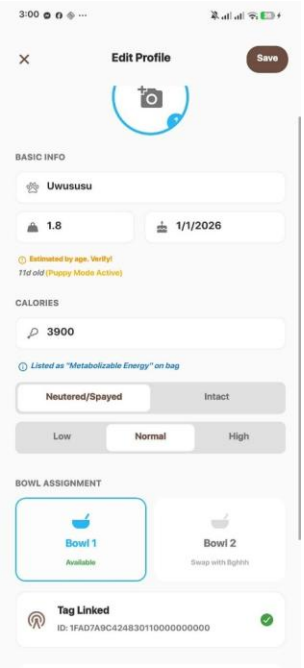

Add New Pet
Pet's Name
e.g., Buddy
Age (months)
Weight (kg)
e.g., 24
e.g., 15
Food Calories (kcal/100g)
Check your dog food bag
Sex / Neuter Status
Neutered/Spayed
Intact
Activity Level
Low
Normal
High
Recommended Daily Portion
0g
per day, based on veterinary formulas.
Save Pet

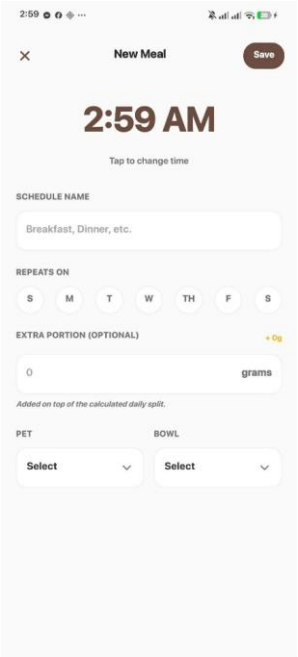

My Pets
Cola

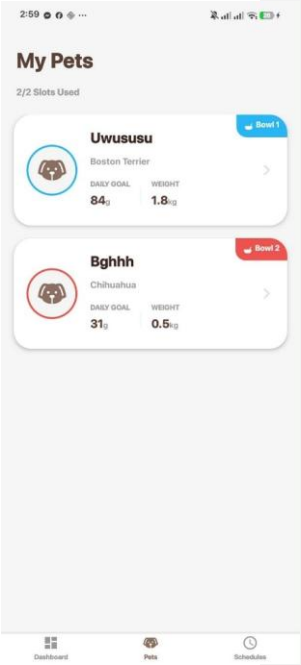

PawFeeds
Bowl 1 - Cola
Menu
Account
Add Device
Reset Device
Manual
Logout
11:44 PM
Cola - Bowl 2
11:44 PM
Cola - Bowl 1

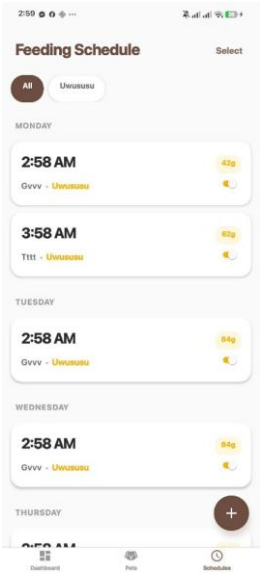

PawFeeds
Bowl 1 - Cola
Food Level
Select Pet to View
Cola
Today's Feedings
Bowl 1
Bowl 2
3:20 AM
Cola - Bowl 2
11:44 PM
Cola - Bowl 2
11:44 PM
Cola - Bowl 1

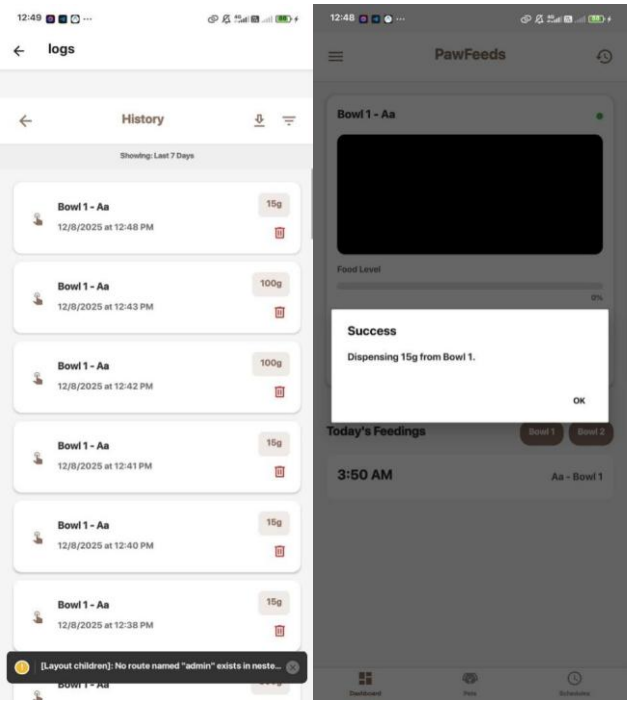

account/account
Account
Email: antonioiggy@gmail.com
Total Pets
0
Total Schedules
0
Change Password
Current Password
Enter current password
New Password
Enter new password
Update Password

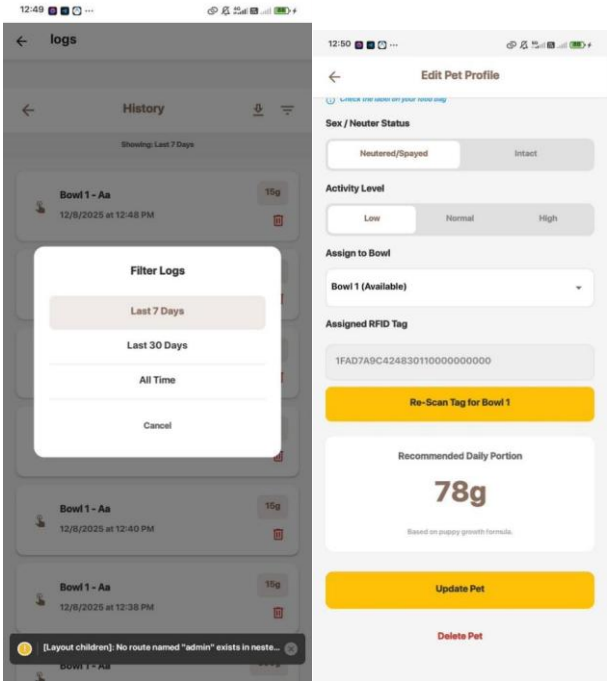

3:00
Edit Profile
Save
BASIC INFO
Uwususu
1.8
1/1/2026
Estimated by age. Verify!
11d old (Puppy Mode Active)
CALORIES
3900
Listed as "Metabolizable Energy" on bag
Neutered/Spayed
Intact
Low
Normal
High
BOWL ASSIGNMENT
Bowl 1
Available
Bowl 2
Swap with Bghhh
Tag Linked
ID: 1FAD7A9C424830110000000000

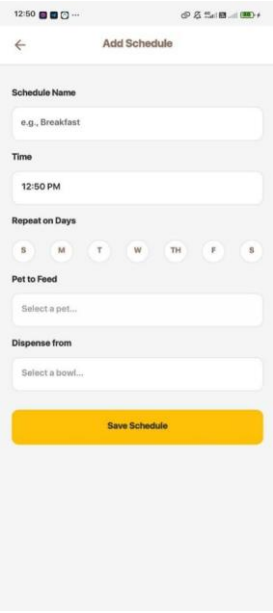

2:59
New Meal
Save
2:59 AM
Tap to change time
SCHEDULE NAME
Breakfast, Dinner, etc.
REPEATS ON
S
M
T
W
TH
F
S
EXTRA PORTION (OPTIONAL)
+ 0g
0
grams
Added on top of the calculated daily split.
PET
BOWL
Select
Select

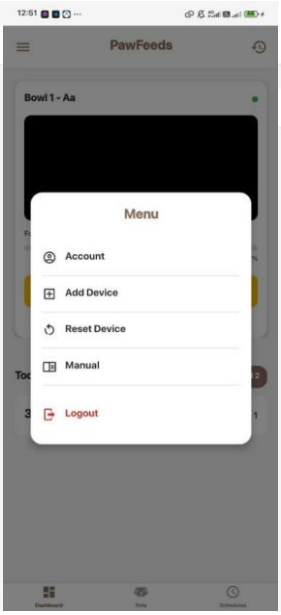

2:59
My Pets
2/2 Slots Used
Bowl 1
Uwususu
Boston Terrier
DAILY GOAL
84g
WEIGHT
1.8kg
Bowl 2
Bghhh
Chihuahua
DAILY GOAL
31g
WEIGHT
0.5kg
Dashboard
Pets
Schedules

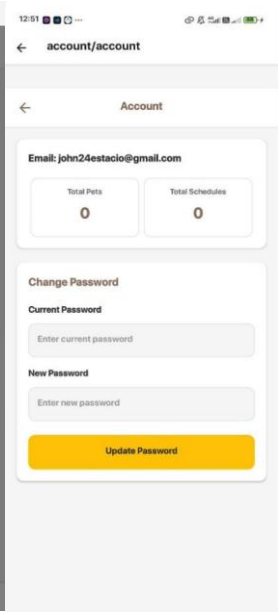

2:59
Feeding Schedule
Select
All
Uwususu
MONDAY
2:58 AM
42g
Gvvv - Uwususu
3:58 AM
62g
Tttt - Uwususu
TUESDAY
2:58 AM
84g
Gvvv - Uwususu
WEDNESDAY
2:58 AM
84g
Gvvv - Uwususu
THURSDAY
Dashboard
Pets
Schedules

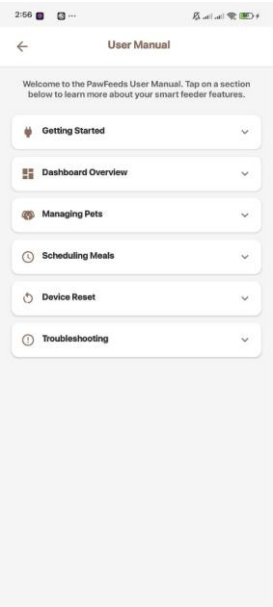

12:49
logs
History
Showing: Last 7 Days
Bowl 1 - Aa
12/8/2025 at 12:48 PM
15g
Bowl 1 - Aa
12/8/2025 at 12:43 PM
100g
Bowl 1 - Aa
12/8/2025 at 12:42 PM
100g
Bowl 1 - Aa
12/8/2025 at 12:41 PM
15g
Bowl 1 - Aa
12/8/2025 at 12:40 PM
15g
Bowl 1 - Aa
12/8/2025 at 12:38 PM
15g
[Layout children]: No route named "admin" exists in neste...
12:48
PawFeeds
Bowl 1 - Aa
Food Level
0%
Success
Dispensing 15g from Bowl 1.
OK
Today's Feedings
Bowl 1
Bowl 2
3:50 AM
Aa - Bowl 1

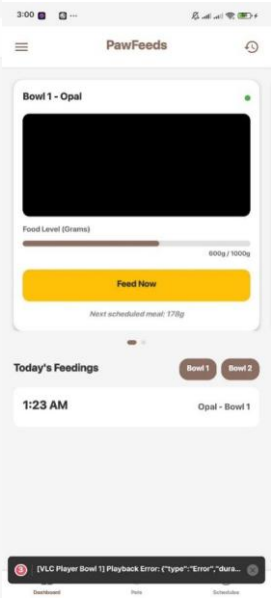

12:49
logs
History
Showing: Last 7 Days
Bowl 1 - Aa
12/8/2025 at 12:48 PM
15g
Filter Logs
Last 7 Days
Last 30 Days
All Time
Cancel
Bowl 1 - Aa
12/8/2025 at 12:40 PM
15g
Bowl 1 - Aa
12/8/2025 at 12:38 PM
15g
[Layout children]: No route named "admin" exists in neste...
12:50
Edit Pet Profile
Sex / Neuter Status
Neutered/Spayed
Intact
Activity Level
Low
Normal
High
Assign to Bowl
Bowl 1 (Available)
Assigned RFID Tag
1FAD7A9C424830110000000000
Re-Scan Tag for Bowl 1
Recommended Daily Portion
78g
Based on puppy growth formula.
Update Pet
Delete Pet

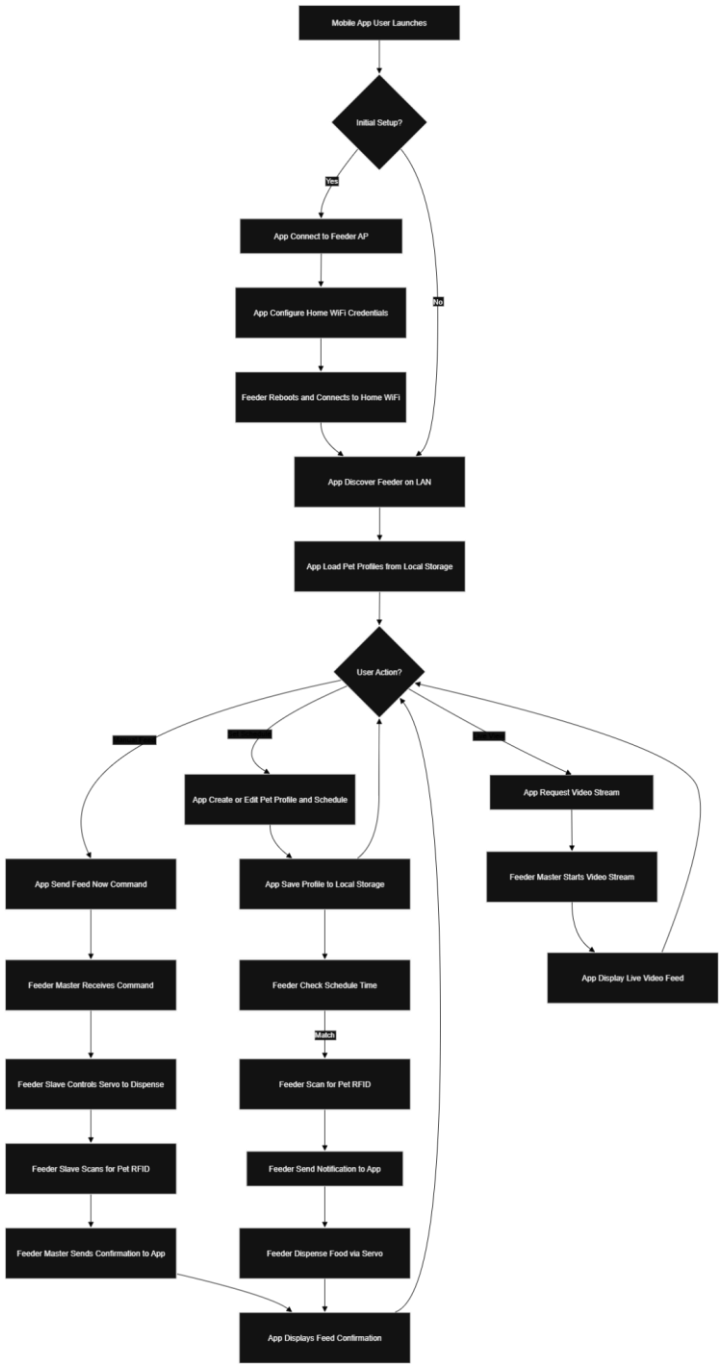

12:50
Add Schedule
Schedule Name
e.g., Breakfast
Time
12:50 PM
Repeat on Days
S M T W TH F S
Pet to Feed
Select a pet...
Dispense from
Select a bowl...
Save Schedule

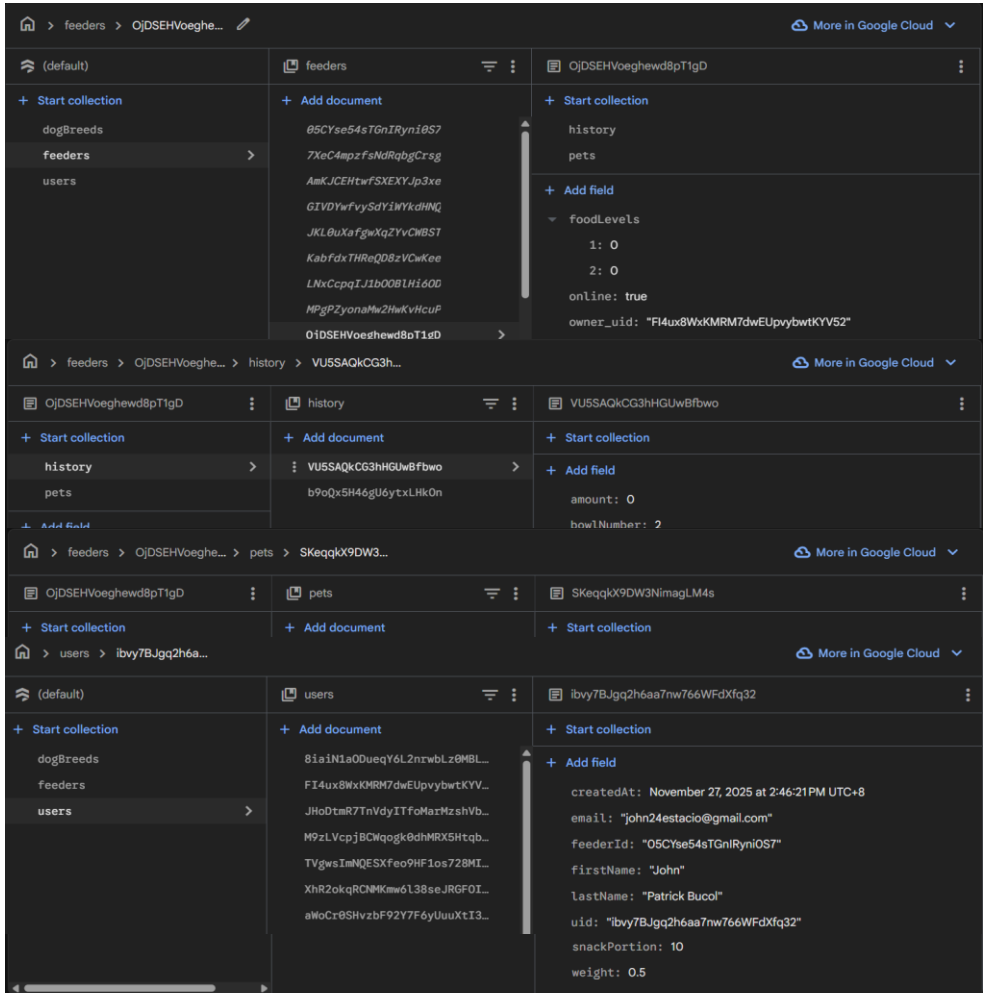

12:51
PawFeeds
Bowl 1 - Aa
Menu
Account
Add Device
Reset Device
Manual
Logout

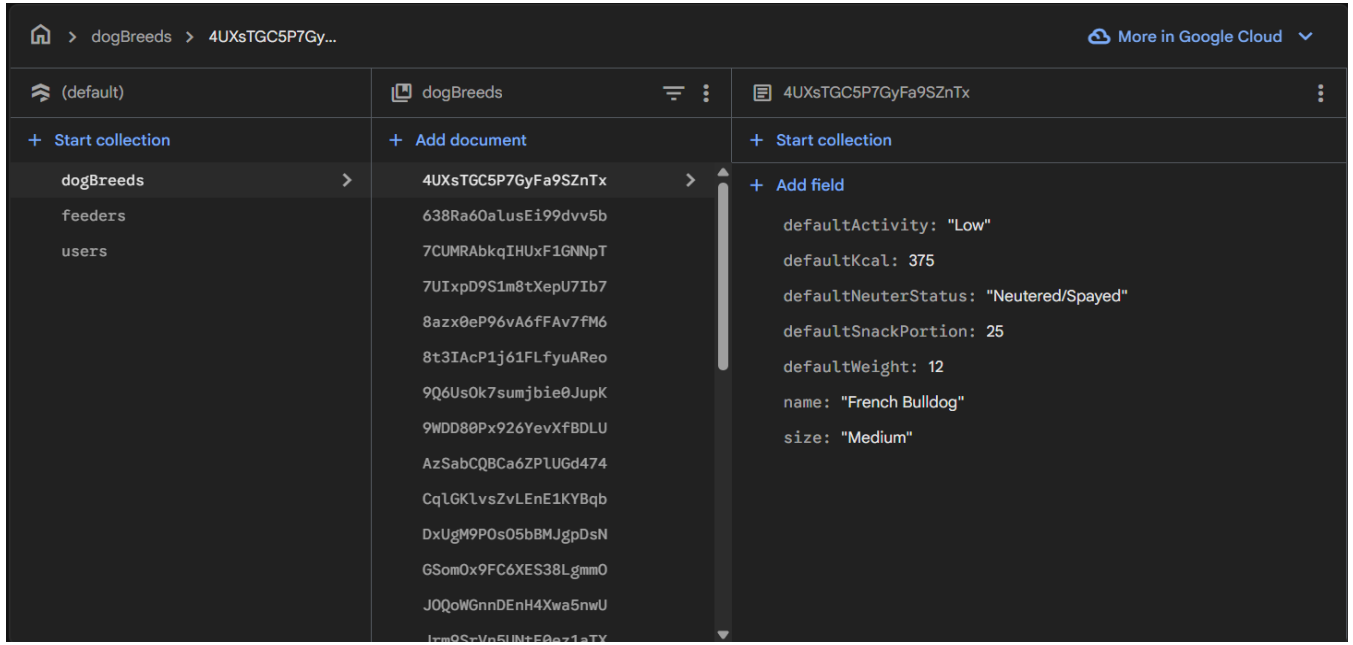

12:51
account/account
Account
Email: john24estacio@gmail.com
Total Pets
0
Total Schedules
0
Change Password
Current Password
Enter current password
New Password
Enter new password
Update Password


2:56
User Manual
Welcome to the PawFeeds User Manual. Tap on a section below to learn more about your smart feeder features.
Getting Started
Dashboard Overview
Managing Pets
Scheduling Meals
Device Reset
Troubleshooting


3:00
PawFeeds
Bowl 1 - Opal
Food Level (Grams)
600g / 1000g
Feed Now
Next scheduled meal: 178g
Today's Feedings
Bowl 1
Bowl 2
1:23 AM
Opal - Bowl 1
[VLC Player Bowl 1] Playback Error: {"type":"Error","dura...


Mobile App User Launches
Initial Setup?
Yes
No
App Connect to Feeder AP
App Configure Home WiFi Credentials
Feeder Reboots and Connects to Home WiFi
App Discover Feeder on LAN
App Load Pet Profiles from Local Storage
User Action?
App Create or Edit Pet Profile and Schedule
App Request Video Stream
App Send Feed Now Command
App Save Profile to Local Storage
Feeder Master Starts Video Stream
Feeder Master Receives Command
Feeder Check Schedule Time
App Display Live Video Feed
Match
Feeder Slave Controls Servo to Dispense
Feeder Scan for Pet RFID
Feeder Slave Scans for Pet RFID
Feeder Send Notification to App
Feeder Master Sends Confirmation to App
Feeder Dispense Food via Servo
App Displays Feed Confirmation


feeders > OjDSEHVoeghe...
More in Google Cloud
(default)
Start collection
dogBreeds
feeders
users
feeders
Add document
05CYse54sTGnIRyni0S7
7XeC4mpzfsNdRqbgCrsg
AmKJCEHtwfSXEXYJp3xe
GIVDYwfvySdYiWYkdHNQ
JKL0uXafgwXqZYvCWBST
KabfdxTHReQD8zVCwKee
LNxCcpqIJ1bOO8lHi6OD
MPgPZyonaMw2HwKvHcuP
OjDSEHVoeghewd8pT1gD
OjDSEHVoeghewd8pT1gD
Start collection
history
pets
Add field
foodLevels
1: 0
2: 0
online: true
owner_uid: "FI4ux8WxKMRM7dwEUpvybwtKYV52"
feeders > OjDSEHVoeghe... > history > VU5SAQkCG3h...
More in Google Cloud
OjDSEHVoeghewd8pT1gD
Start collection
history
pets
history
Add document
VU5SAQkCG3hHGUwBfbwo
b9oQx5H46gU6ytxLHkOn
VU5SAQkCG3hHGUwBfbwo
Start collection
Add field
amount: 0
bowlNumber: 2
feeders > OjDSEHVoeghe... > pets > SKeqqkX9DW3...
More in Google Cloud
OjDSEHVoeghewd8pT1gD
Start collection
pets
Add document
SKeqqkX9DW3NimagLM4s
Start collection
users > ibvy7BJgq2h6a...
More in Google Cloud
(default)
Start collection
dogBreeds
feeders
users
users
Add document
8iaiN1aODueqY6L2nrwbLz0MBL...
FI4ux8WxKMRM7dwEUpvybwtKYV...
JHoDtmR7TnVdyITfoMarMzshVb...
M9zLVcpjBCWqogk0dhMRX5Htqb...
TVgwsImNQESXfeo9HF1os728MI...
XhR2okqRCNMKmw6l38seJRGFOI...
aWoCr0SHvzbF92Y7F6yUuuXtI3...
ibvy7BJgq2h6aa7nw766WFdXfq32
Start collection
Add field
createdAt: November 27, 2025 at 2:46:21 PM UTC+8
email: "john24estacio@gmail.com"
feederId: "05CYse54sTGnIRyni0S7"
firstName: "John"
lastName: "Patrick Bucol"
uid: "ibvy7BJgq2h6aa7nw766WFdXfq32"
snackPortion: 10
weight: 0.5


dogBreeds > 4UXsTGC5P7Gy...
More in Google Cloud
(default)
dogBreeds
4UXsTGC5P7GyFa9SZnTx
+ Start collection
+ Add document
+ Start collection
dogBreeds
feeders
users
4UXsTGC5P7GyFa9SZnTx
638Ra6OalusEi99dvv5b
7CUMRAbkqIHUxF1GNNpT
7UIxpD9S1m8tXepU7Ib7
8azx0eP96vA6fFAv7fM6
8t3IAcP1j61FLfyuAReo
9Q6UsOk7sumjbie0JupK
9WDD80Px926YevXfBDLU
AzSabCQBCa6ZPlUGd474
CqlGKlvsZvLEnE1KYBqb
DxUgM9POsO5bBMJgpDsN
GSomOx9FC6XES38LgmmO
JOQoWGnnDEnH4Xwa5nwU
+ Add field
defaultActivity: "Low"
defaultKcal: 375
defaultNeuterStatus: "Neutered/Spayed"
defaultSnackPortion: 25
defaultWeight: 12
name: "French Bulldog"
size: "Medium"

Appendix C: Evaluation Tool

Name: Date:

Directions:

This questionnaire is design to evaluate the PAWFEEDS system. Please answer each question base on your experience as a dog owner. Part A check the option that best suits you. Part B, rate each statement according to your rating. Using the point scale below:

1- Strongly Disagree
2- Disagree
3- Neutral
4- Agree
5- Strongly Agree

For part C kindly provide your comments, suggestions and opinions in your own words. Your responses will use to improve the system further.

Part A. Survey Questionnaire

1. do you own a dog/s?

_Yes _No

2. how many dogs you own?

_2 _3 _(more than 3)

3. what type/s of food do you usually feed your dog/s?

_Dry Food _Wet Food _Both

4. How many times a day do you feed your dog/s?

_Once _ Twice _Thrice or more

5. are your dogs a picky eater?

_Yes _No _Sometimes

Part B: Evaluation

Statement	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The system makes it easier to follow consistent feeding schedules.					
The mobile application is easy to navigate.					
Adjusting input feeding times or portion sizes through the app is convenient.					
The PAWFEEDS system is good and reliable for dog owners.					
The live streaming video is clear and can easily monitor the dispensed food.					
The system dispensed food accurate portion sizes based on the dog/s needs.					
Automated Feeding is better than old traditional feeding.					
The RFID feature is working correctly and recognize correct dog tags.					
I would Recommend this system to other dog owners.					

Part C: Open Ended Questions

1. what do you like mostly about our system?

2. What features would you suggest on improving or adding?

Appendix D: Relevant Source Code

1. Main Feeder Firmware (Hardware)

This is the Master.txt firmware file that integrates the main ESP32 controller and consists of methods for managing the physical feeder like, triggerDispense and handleRfidRead, and load cell (weight) sensing functions.

```
// Dispenser Management State
struct DispenserState {
  DispenserMode mode;
  Servo* servo;
  int bowlId;
  float targetTotalWeight;
  float lastStableWeight;
  unsigned long lastWeightChangeTime;
  unsigned long stateStartTime;
  int unclogAttempts;
  UnclogStep unclogStep;
};

// Core Dispense Method
void triggerDispense(int bowlNumber, int amount) {
  DispenserState* disp = (bowlNumber == 1) ? &dispenser1 : &dispenser2;
```

```
  HX711* scale = (bowlNumber == 1) ? &loadCellBowl1 : &loadCellBowl2;

  if (disp->mode != D_IDLE) {
    Serial.printf("[SERVO] Bowl %d is already busy. Command ignored.\n",
bowlNumber);
    return;
  }

  float currentWeight = getWeightInGrams(*scale, bowlNumber);
  disp->targetTotalWeight = currentWeight + amount;
  disp->lastStableWeight = currentWeight;
  disp->lastWeightChangeTime = millis();
  disp->unclogAttempts = 0;
  disp->mode = D_DISPENSING;
  disp->servo->write(SERVO_FORWARD);
}
```

2. Camera Firmware (Hardware)

This is the Slave.txt camera firmware file that integrates the ESP32-CAM module and consists of methods for processing video feeds like, startCameraServer and stream_handler, and IP synchronization functions.

```
// Video Streaming Handler
```

```
esp_err_t stream_handler(httpd_req_t *req){
  camera_fb_t * fb = NULL;
  esp_err_t res = ESP_OK;
  size_t _jpg_buf_len = 0;
  uint8_t * _jpg_buf = NULL;
  char * part_buf[64];

  res = httpd_resp_set_type(req, _STREAM_CONTENT_TYPE);
  if (res != ESP_OK) return res;

  while (true) {
    fb = esp_camera_fb_get();
    if (!fb) {
      res = ESP_FAIL;
      break;
    }

    _jpg_buf_len = fb->len;
    _jpg_buf = fb->buf;

    // Send multipart jpeg frames
    if (res == ESP_OK) res = httpd_resp_send_chunk(req,
_STREAM_BOUNDARY, strlen(_STREAM_BOUNDARY));
```

```
    if (res == ESP_OK) {
      size_t hlen = snprintf((char *)part_buf, 64, _STREAM_PART, _jpg_buf_len);
      res = httpd_resp_send_chunk(req, (const char *)part_buf, hlen);
    }
    if (res == ESP_OK) res = httpd_resp_send_chunk(req, (const char *)_jpg_buf,
_jpg_buf_len);
    esp_camera_fb_return(fb);
    if (res != ESP_OK) break;

    delay(1); // 1ms delay for high FPS
  }
  return res;
}
```

3. Firebase Backend Functions (Software)

This is the index.ts backend service file that integrates the Firebase serverless environment and consists of methods for managing schedules and alerts like, scheduledFeedChecker and onFeederError, and push notification delivery functions.

```
// structure for Schedule data
interface ScheduleData {
  time: string;
```

```
  repeatDays: string[];
  petId: string;
  bowlNumber: number;
  portionGrams: number;
  isEnabled: boolean;
}

// Scheduled Feed Checker
export const scheduledFeedChecker = onSchedule(
  {
    schedule: "every 1 minutes",
    timeZone: "Asia/Singapore",
    region: "asia-southeast1",
  },
  async (event: ScheduledEvent) => {
    logger.info("Running scheduled feed checker...");
    // time parsing logic

    const feedersSnapshot = await firestore.collection("feeders").get();
    for (const feederDoc of feedersSnapshot.docs) {
      const schedulesRef = feederDoc.ref.collection("schedules");
      const q = schedulesRef.where("isEnabled", "==", true);
```

```
      q.get().then(async (scheduleSnapshot) => {
        for (const scheduleDoc of scheduleSnapshot.docs) {
          const schedule = scheduleDoc.data() as ScheduleData;

          if (schedule.time === currentTime &&
schedule.repeatDays.includes(currentDay)) {
            // Dispatch RTDB await_rfid command and trigger notification
          }
        }
      });
    }
  }
);
```

4. Video Relay Server (Software)

This is the relayserver_hls.js video relay file that integrates FFmpeg and consists of methods for processing live camera streams like, startStreamProcess, and RTMP network handling functions.

```
// Creates and starts a persistent ffmpeg
function startStreamProcess(streamKey, cameraUrl) {
  const outputUrl = `${RTMP_BASE_URL}/${streamKey}`;
```

```
  const ffmpegProcess = ffmpeg(cameraUrl)
    .inputFormat('mjpeg')
    .addInputOptions(FFMPEG_INPUT_OPTIONS)
    .videoFilters('setpts=PTS-STARTPTS') // Reset timestamps to 0 to prevent
sync issues
    .addInputOption('-analyzeduration 500000')
    .addInputOption('-probesize 500000')
    .addOptions(FFMPEG_OUTPUT_OPTIONS)
    .output(outputUrl)
    .on('start', (commandLine) => {
      console.log(`[FFmpeg ${streamKey}] Active.`);
    })
    .on('error', (err, stdout, stderr) => {
      console.log(`[FFmpeg ${streamKey}] Reconnecting in 5 seconds...`);
      setTimeout(() => startStreamProcess(streamKey, cameraUrl), 5000);
    })
    .on('end', () => {
      setTimeout(() => startStreamProcess(streamKey, cameraUrl), 1000);
    });

  ffmpegProcess.run();
}
```

Appendix E: Ethical Clearance

Appendix F: Plagiarism Report

Appendix G: Research Critique and Editing Certificate

Appendix H: Curriculum Vitae